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Practical Intelligence: Generative AI Toolkit for Nurse Education

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Practical Intelligence: Generative AI Toolkit for Nurse Education

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Toolkit Developed: July 25, 2025
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Part One: Understanding Generative AI

Introduction to Generative AI

Generative AI (Artificial Intelligence) is a class of tools capable of producing text, images, and even code based on prompts entered by the user. Generative AI platforms, such as ChatGPT, Gemini, and Claude, are becoming increasingly accessible to educators, students, and clinical professionals. Generative AI is undergoing rapid development, and educators are adopting generative AI into their daily workflows to draft emails, revise policy documents, and brainstorm learning activities. Though these uses are not inherently problematic, without a grounded understanding of the model's mechanics, nurse educators risk misapplying AI tools in ways that compromise academic and professional integrity.

This toolkit is designed to serve as a foundational introduction for nurse educators, regardless of prior experience with artificial intelligence. As faculty, researchers, and clinicians, nurse educators are well-positioned to use AI tools with discernment if the underlying technologies and their limitations are clearly understood.

What is Generative AI?

Generative AI refers to a class of machine learning systems trained to produce content such as text, images, or audio. The models are created by learning from patterns in vast amounts of data. In contrast to traditional rule-based systems that follow fixed instructions, generative models use statistical prediction to determine the most likely next word or image fragment based on a prompt.

At its core, generative AI is built around probabilistic modeling. The goal of the system is to generate the most statistically likely sequence of words or visual elements based on prior examples. A generative model like ChatGPT are trained on hundreds of billions of words from books, websites, and other publicly available sources. This training allows it to calculate the next most likely word in a sentence by learning the probabilities of word sequences. For example, when prompted with “The nurse assessed the newborn’s…,” the model may predict “heart rate,” “reflexes,” or “weight” based on how often these words follow that phrase in its training data.

AI models do not think or reason in the human sense. Instead, they rely on a process called tokenization, which breaks text into chunks called “tokens”. Tokens may be part of words, whole words, or phrases. To write, the model selects the most probable next token based on patterns it has previously encountered. This allows it to generate fluent, coherent-sounding responses that may be inaccurate.

When you type a prompt into a generative AI tool, your prompt is broken into tokens. The system uses your tokens to query its trained model to select the first token based on which token is the most probable reply. Asking the same question in different words will result in different tokenization of your question and give you different responses. Each token selected by the AI tool affects the probability of the next token in the sequence, continuing token by token until a full response is formed. Each generative AI tool uses a different probability weighting and different set of tokens, so asking the same question to different AI tools can result in different answers.

Writing by identifying the next most probable token makes AI-generated writing seem fluent and intelligent, even when it produces text that is wrong, misleading, or nonsensical. Importantly, the training data that informs these models often includes outdated information, non-peer-reviewed content, and sometimes misinformation from the internet. This is especially relevant for nurse educators who rely on current, evidence-based resources in their academic and clinical work because the surface fluency of AI-generated text can mask significant factual or conceptual errors.

A key point to remember is that generative AI lacks a sense of meaning. Humans write with intention, purpose, and a mental model of their audience. In contrast, AI tools generate content based on word placement structure. The model is not aware that a "nursing care plan" must follow logical clinical reasoning. It simply knows that these words often appear together. Unlike a nurse educator, generative AI cannot apply professional judgment. Even more troubling is how these models can confidently produce fabricated citations when simulating academic writing. When prompted to include references, the system may produce realistic-looking journal citations that do not exist because the model has learned how references are structured.

The implication is clear: educators cannot outsource truth evaluation to a generative AI model. Verification and critical thinking remain essential.

Why Generative AI Fails

Because generative AI models rely on pattern recognition rather than understanding, their failures follow predictable patterns. These include:

- Hallucination of facts, where the model confidently generates incorrect or fabricated information that sounds plausible.
- Misinterpretation of nuance, such as misunderstanding clinical context, tone, or ethical implications in sensitive discussions (e.g. patient safety, trauma-informed care).

- Reinforcement of bias, because AI can reflect or amplify inequities present in training data, affecting decision-making or recommendations in teaching and clinical simulation scenarios.
- Inability to verify truth, if AI lacks access to real-time validation, leading to outdated or unverifiable responses when accuracy is essential.
- Surface-level reasoning, where the model provides generic or shallow responses that lack depth or critical analysis, which is problematic in academic or evidence-based nursing environments.

Predictive, Not Epistemic Reasoning

Generative AI models function by identifying the most statistically likely next word or phrase, not by assessing the factual accuracy or evidentiary strength of that phrase. This is a fundamental limitation. These models are optimized for fluency, not truth. While they can produce grammatically correct and seemingly plausible responses, they do not verify facts, cross-check data, or access the most current evidence unless specifically augmented with external tools.

For example, a nurse educator might prompt "explain the difference between formative and summative assessment in nursing education." The model may generate a fluent explanation and incorrectly state that "formative assessments are used at the end of the course to determine whether learning objectives have been met," reversing the core definitions. Because the model selects the most probable sentence structure rather than evaluating correctness, it can confidently present incorrect academic content, potentially misleading educators or students.

Contextual Misplacement

AI systems can place the correct term or concept in the wrong context. This occurs because the model does not understand the meaning of what it generates, it recognizes token patterns, not concepts. Thus, even when it uses familiar terms, it can misapply them in ways that may go unnoticed by non-experts but are critical in clinical or academic environments.

Consider an educator requesting "a discussion question about nurse delegation policies in long-term care." The AI may generate: "How does the nurse practitioner delegate medication administration to a registered nurse in the home setting?" This example misplaces the concept of delegation by assuming that registered nurses' medication administration authority is delegated by nurse practitioners. The question appears logical but teaches incorrect information about scope of practice.

Overgeneralization from Biased Training Data

Generative AI models are trained on internet-based texts, public datasets, and open-source documents. Because of this, they tend to reflect dominant narratives, popular opinion, and can include outdated practices. These data sources can underrepresent evidence from peer-reviewed nursing literature or guidelines from authoritative bodies that are generally housed behind paywalls. This bias toward common, not credible, information can mislead users into accepting generalized but inaccurate content.

For instance, a nurse educator asking for "a short paragraph explaining informed consent in childbirth" might receive a response stating: "Informed consent is given during the first prenatal visit and allows the physician to proceed with necessary procedures during labor." This inaccurate, oversimplified, and physician-centric explanation ignores shared decision-making, situational consent, and the midwifery model of care. The model generalizes from more common web narratives, which may lack a nursing perspective or a patient-centered framework.

Chain of Reasoning Failures

When AI models are asked to generate responses that require multiple logical steps, such as building a teaching plan or evaluating a complex case, they can go off track. Errors introduced early in the chain compound into more serious problems later. This is especially concerning when the user expects the AI to reason through clinical or ethical dilemmas.

For instance, a nurse educator might prompt the AI: "Create an outline, three teaching points, and two discussion questions for a lecture on the stages of labor." The AI might produce an outline that incorrectly identifies "effacement" and "station" as stages of labor. Based on that flawed outline, it could then suggest incorrect teaching points like "Describe the fifth stage of labor: cervical positioning." It could then go on to generate discussion questions that further build on the error. The foundational errors shape the subsequent content around an inaccurate premise from the first step. If used without review, such materials can propagate misunderstandings in student learning or clinical practice.

Ungrounded Prompts Yield Hallucinations

AI systems "hallucinate" content when prompts are too vague or under-specified. Without clear parameters, the model improvises details to fill in gaps, even if those details are factually incorrect. Because these inventions are smoothly woven into the narrative, they may be mistaken for legitimate content unless the user has domain-specific knowledge.

For example, an educator might ask: "Write a paragraph explaining ethical issues with fetal monitoring." The AI may return a paragraph stating that "external fetal monitoring is illegal in some U.S. states due to privacy concerns," a fabricated claim that appears plausible but is entirely untrue. The hallucination results from the model blending ethics and legality in a way that makes statistical sense within its training corpus, but has no real-world foundation. In nursing education, such hallucinations can confuse students or propagate misinformation in classroom settings if not critically appraised.

Review: Red Flags for AI-Generated Text

- Citation does not exist or is incorrectly attributed
- Clinical content is plausible but outdated
- Language is generic or lacks disciplinary specificity
- Arguments lack logical coherence or completeness
- AI-generated images or examples misplace key elements
- Quoted policies or guidelines are not verifiable
- Output sounds confident but fails expert review

Implications for Nurse Educators

The growing availability of generative AI presents both opportunities and risks for nurse educators. While these tools can save time and spark creativity, they do not substitute for professional expertise, evidence-based practice, or peer review.

Educators can use AI-generated content to draft syllabus descriptions, simulate student responses, or create teaching materials. However, without careful validation, this content may reflect errors, oversimplifications, or biases that mislead students or misrepresent disciplinary standards. Ultimately, nurse educators play a critical role in maintaining the academic rigor, clinical accuracy, and ethical foundation of nursing education.

History of Artificial Intelligence in Healthcare

Before AI had a name in healthcare, we were already using algorithms. In the 1960s and 70s, hospitals like Massachusetts General and LDS Hospital were developing rule-based systems designed to aid clinical decisions. These systems had limited stand-alone services, for example some managed lab reporting or billing. An example of these early systems is the COSTAR system at Massachusetts General Hospital.[1]

COSTAR stands for Computer Stored Ambulatory Record. This system was developed in the late 1960s in collaboration with computer scientists at Harvard University. What was unique about this system was that it was focused on outpatient and primary care settings, unlike most systems at the time which focused on inpatient care. COSTAR used a problem-oriented medical record system, which means it organized the record by the patient's problem list. This was a major innovation at the time. It also provided alerts and reminders for preventive services and drug interactions; an example of real-time alerts built from simple logic rule-based algorithms. This system was adopted and adapted by multiple health centers, particularly in academic settings, through the 1980s and 1990s. COSTAR's source code was made publicly available (open source before we had open source). This allowed open development and experimentation which is not available in today's proprietary electronic health records.

Other computer systems used logic rules to flag abnormal lab values or guide antibiotic choices. One of the most famous was MYCIN.[2] Developed at Stanford University in the early 1970s, MYCIN was built to support clinical decision-making for bacterial infections. Its primary goal was to support an infectious disease diagnosis and recommend appropriate antibiotic therapy.

Like COSTAR, MYCIN used rule-based logic. This means the system was built on "if-then" statements to decide what to report. A breakthrough was that MYCIN could manage clinical ambiguity. This meant it could still complete its algorithmic reasoning even when there was missing data. It could also provide a

[1] Hattwick M. A. W. (1979). Computer Stored Ambulatory Record Systems in Real Life Practice. Proceedings of the Annual Symposium on Computer Application in Medical Care, 761–764.
[2] Buchanan, B.G. & Shortliffe, E.H. (1984). Rule Based Expert Systems: The MYCIN Experiments of the Stanford Heuristic Programming Project. Addison-Wesley Longman Publishing Co., Inc. Boston, MA.

justification for its recommendations by producing the rule-chain, an early example of transparent reasoning.

MYCIN was never deployed in live clinical environments due to barriers that are still relevant today. First, physicians didn't have clear guidance on how the output affected their responsibility and the hospitals didn't want legal and ethical uncertainty. MYCIN required specialized terminals where physicians would re-enter information from the main computer system into MYCIN. This didn't fit the workflow of an ICU and is an early example of the problems that occur when systems are not interoperable. Despite these flaws, MYCIN was one of the first systems to show that computers could mimic clinical reasoning. Some of the concepts used to create MYCIN informed the development of modern clinical decision support systems.

In the early 1980s, the APACHE system was used to measure the severity of illness for patients admitted to intensive care.[3] APACHE stands for Acute Physiology and Chronic Health Evaluation. The goal of the program was to objectively quantify illness severity using readily available physiological and clinical variables. APACHE gave each patient a score between 0 and 70, with higher scores indicating greater illness severity. It could predict the ICU mortality risk based on the patient condition at admission, and was used to support resource allocation, benchmarking, and quality assurance.

Before APACHE, there was no standardized, data-driven method to assess the baseline severity of illness among ICU patients. This meant you couldn't compare the outcomes across institutions or patient populations. APACHE became the global standard for ICU severity scoring and remains one of the most cited and respected ICU scoring tools. This early model has driven advancements in predictive analytics, AI-based mortality prediction, and quality monitoring.

During the next phase of development, hospitals began embedding more sophisticated tools within electronic health records. One notable example was the St. John's Sepsis Agent, deployed through Cerner.[4] It monitored real-time vitals and lab data to detect sepsis risk. This was a clear shift from passive data

[3] Wagner, D. P., & Draper, E. A. (1984). Acute physiology and chronic health evaluation (APACHE II) and Medicare reimbursement. Health care financing review, Suppl(Suppl), 91–105.
[4] Wawrose R, Baraniuk M, Standiford L, Wade C, Holcomb J, Moore L. Comparison of Sepsis Screening Tools' Ability to Detect Sepsis Accurately. Surg Infect (Larchmt). 2016;17(5):525-529. doi:10.1089/sur.2015.069

storage to active clinical surveillance. It scanned patient records, including a 30-hour look-back window to identify abnormal vital signs or laboratory values that could indicate sepsis. It integrated evidence-based criteria such as the CMS sepsis guidelines. Hospitals were able to customize the alerts to better fit their clinical workflow. This system enhanced early detection, improved predictive accuracy, and supported clinical decision-making.

From Rule-Based Systems to Generative Models

Early decision-support tools weren't machine learning, but they marked the beginning of embedded intelligence in patient care. Rule-based models, like the ones used in COSTAR, MYCIN, and the St. John's Sepsis Agent depend on explicitly programmed logic. These are "if-then" systems written by humans. They're highly transparent. But these models only work as well as the rules we give them. They're brittle when patient presentations don't fit the anticipated mold.

This is where machine learning steps in. Unlike rule-based models, machine learning systems don't rely on predefined rules. Instead, they learn from data. A machine learning model might analyze tens of thousands of patient records to discover patterns that predict sepsis or deterioration. What makes this unique is that the computer can detect patterns a human might not notice.

But the power to detect hidden patterns comes at a cost. Machine learning models can be harder to interpret. Their decision pathways are statistical, not explicitly programmed. This makes them harder to audit, harder to explain, and in some settings, harder to trust.

Deep learning takes machine learning a step further. It's a subtype of machine learning that uses neural networks. Neural networks are layered structures modeled loosely on how neurons work in the brain. Deep learning is especially powerful with unstructured data, like radiology images, pathology slides, clinical notes, or audio recordings. It doesn't need a human to define what features to look for. Instead, it learns those features on its own.

These models don't just identify known patterns, which we could find using if-then algorithms. Machine learning and deep learning detect new and unknown patterns. But traditional deep learning is still mostly discriminative: it predicts or classifies based on input data. This makes machine learning tools useful for a limited set of clinical tasks.

Epic partnered with IBM Watson to explore a bold idea: could advanced computing help clinicians?[5] It sparked a turning point. By 2020, Epic and Cerner both had embedded machine learning models.[6] These were designed to predict sepsis, mortality, and readmission risk using data already captured in the EHR.

The Rise of Generative AI

The next evolution of computer reasoning took a bold step forward, building computer models that generate. Rather than just assessing a scan or summarizing a note, these systems can create synthetic images, write documentation, and even simulate conversation. Unlike prior computer reasoning models that classified or scored data, generative models learn the structure of a dataset well enough to create new examples. A model trained on radiology scans might generate synthetic images. A model trained on clinical notes might draft synthetic documentation.

The main innovation for language models was transformers, introduced in 2017 by Vaswani and colleagues.[7] Transformers turn words into tokens as the basic units of meaning to analyze and generate text with nuance. A token may be a word, a part of a word, or a group of words. Transformers allowed the computer to identifying how each token was related to the other tokens in a piece of writing. This allowed the computer to assign weight to the relationship between a token and all the other tokens used in a piece of writing. Using that weight, the tool can predict the next token in a sequence based on the weighted probability. This breakthrough is foundational to how modern AI systems understand and generate language. This is how tools like ChatGPT and Med-PaLM simulate context-aware content.

From this architecture came Large Language Models, or LLMs. These models are trained on vast datasets and can perform a wide range of language-based

[5] Monegain, B. (May 5, 2015). Epic, Watson at work on interoperability. Healthcare IT News. https://www.healthcareitnews.com/news/epic-watson-work-interoperability
[6] Drees, J. (May 13, 2020). AI for EHRs: 4 updates on Epic, Cerner, Allscripts & Meditech capabilities. https://www.beckershospitalreview.com/healthcare-information-technology/ehrs/ai-for-ehrs-4-updates-on-epic-cerner-allscripts-meditech-capabilities/
[7] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17). Curran Associates Inc., Red Hook, NY, USA, 6000–6010.

predictive tasks. With enough data, they generalize effectively enough to summarize notes, draft writing, and even simulate conversations.

Generative language tools are now being integrated into decision support systems, discharge planning workflows, and education platforms. This means we're no longer asking if AI will be part of clinical care. It already is. Instead, nurse leaders are asking how to direct the use of AI systems to ensure patient-centered care.

The Role of Nurse Leaders

Today, nurses interact with AI as a part of patient care. This comes in the form of pre-screening tools, chat bots, risk scores, and smart documentation tools. While these tools can support safer, more efficient care, they also raise important challenges. For example, nurses need to be aware that data bias in the training models can result in systematic inequities in the AI output. We need to determine how we will evaluate opaque models that cannot be independently evaluated. We don't know how effective these tools can be once nurses experience alert fatigue from constant prompts. This is why nurse leadership is essential in the adoption of AI in healthcare.

As a nurse educator, your role isn't just to use these systems. Your role is to question them. Who trained the model? What populations were represented in the data? Is the model reinforcing disparities? Does it enhance or undermine clinical reasoning? These are clinical, ethical, and organizational questions that require the kind of system thinking and frontline knowledge that nurse leaders bring. The next generation of AI tools will be more embedded, more conversational, and more personalized. But no matter how sophisticated they become, they never replace human judgment. Nursing expertise in patient context, workflow integration, and ethical decision-making is more important than ever.

Is AI the Right Tool for the Task?

One of the primary benefits of generative AI is its ability to support drafting, brainstorming, and formatting tasks, potentially increasing productivity in teaching, research, and practice. Yet, actual productivity gains with AI have been small. Only 20% of users report AI saves them at least four hours of work per week.[8] Even simple and widely adapted tools like chatbots result in modest productivity gains.[9] This raises a critical question for educators: What uses of AI help me be more productive?

This chapter offers a decision-making framework that encourages nurse educators to critically assess the fit between AI capabilities and task demands. Rather than relying on assumptions about technological efficiency, nurse educators can make informed, ethical, and context-sensitive choices about when and how AI tools are appropriate. As nurse educators prepare students for practice in increasingly complex, technology-rich environments, it is essential that we first develop the judgment to use those technologies wisely ourselves.

Tolerance for error

In healthcare and nursing education, error can compromise learning, propagate misinformation, erode trust, and in some cases, contribute to unsafe clinical thinking. Nurse educators have long served as critical filters between knowledge and application. When using AI tools, this responsibility remains: educators evaluate not just the usefulness of AI, but its potential to mislead, misrepresent, or reinforce harm. A fluent but flawed explanation generated by AI may be difficult to detect without deep content expertise, and once adopted into a syllabus, student assignment, or published article, it can persist unchallenged.

For nurses, tolerance for error can be based on the level of harm that may occur from the output. Tolerance for error determines the extent of human review needed when AI tools are used, when the tolerance for error is low the amount

[8] Alexander Bick, Adam Blandin and David Deming, "The Impact of Generative AI on Work Productivity," St. Louis Fed On the Economy, Feb. 27, 2025.
[9] Humlum, Anders and Vestergaard, Emilie, Large Language Models, Small Labor Market Effects (April 15, 2025). University of Chicago, Becker Friedman Institute for Economics Working Paper No. 2025-56, Available at SSRN: https://ssrn.com/abstract=5219933 or http://dx.doi.org/10.2139/ssrn.5219933

of editing and validation of output increases. When evaluating tolerance for error, nurse educators can ask themselves these questions:

- How harmful would it be if there were an error in the output?
- Could there be harm due to bias or stereotype reinforcement that may harm students or patients?
- Could it introduce legal or ethical inaccuracies?
- Will others rely on the content without revision (e.g., students, committee members, public)?

Even when AI-generated text sounds appropriate, it must be interrogated for bias, oversimplification, and deviation from professional values. In high-risk, high-impact, or high-sensitivity tasks, the risk of harm outweighs the convenience of AI.

Novelty

Because generative AI uses patterns in its training data to predict the most probable next word or image segment, it performs better at routine or common tasks. Tasks like summarizing standard documents, formatting reports, and generating emails will be well-represented patterns in the training data. Routine tasks have clear expectations and match common structures already found in public data sets like meeting minutes and lesson plans. These tasks align well with generative AI's strengths in recognizing and reproducing patterns based on their training data.

Using generative AI for routine tasks, such as creating a template for a medical history, is likely to produce output that requires minimal editing. Similarly, generative AI will likely do a good job outlining frameworks for implementing a quality improvement project. Both tasks rely on information and reporting formats that are widely available in public data, and we can assume societal knowledge will be integrated into the models training.

Generative AI is capable of producing highly creative output, particularly when remixing known ideas, formats, or examples in novel combinations. When prompted to engage with tasks that are truly novel, meaning those that lack clear precedent, structure, or anchoring data, the model may generate content that is plausible in tone but conceptually flawed. This tendency arises from the model's objective to produce content based on next token probability. As a result, it may misapply frameworks, overgeneralize concepts, or fabricate details in an effort to fulfill the prompt. Educators who are aware of this tendency can use generative AI for early-stage brainstorming, particularly when the goal is to surface a wide range of ideas quickly. When using AI in this way, be prepared to encounter and discard suggestions that are irrelevant, unusable, or inappropriate for the context.

User Capacity and Context

Improving productivity with generative AI depends on the user's capacity to guide the tool, assess the output, and align the result with professional standards. Educator experience, institutional policies, access to secure platforms, and awareness of AI's limits all play a critical role in determining whether AI is truly a good fit.

Educators who develop skill in AI use and integrate it thoughtfully into familiar workflows can gain substantial value from the tool. Without skillful prompting and thoughtful integration, educators may find AI a frustrating editing burden. In this sense, AI effectiveness is not fixed. Effectiveness evolves with the user's expertise, judgment, and the learning environment. With experience, educators can build prompts that have clarity and specificity. Educators who are new to AI may overgeneralize their prompts or assume the tool "understands" more than it does.

Educators can assume that AI outputs always require review. This includes checking references, ensuring terminology is up-to-date, and confirming alignment with learning objectives or evidence-based standards. AI can accelerate drafting, but not validation. Educators using AI in high-stakes contexts (e.g., publishing, accreditation reporting, and curriculum development) do well to treat the outputs as provisional and incomplete. Using AI to assist with unfamiliar subject matter increases the time that educators use to validate the information and the risk that educators will not recognize flawed or biased outputs.

Transparency

Transparency, the practice of clearly communicating how decisions were made, is essential in nursing research, scholarship, and teaching. When nurse educators use AI tools as part of their workflow, documenting the ways the AI tools contributed to the work ensures that peer reviewers, readers, and students can critically evaluate the rigor and validity of the work. Being transparent about your AI use requires you to keep a record of your interactions with AI. For some projects, this may be as simple as sharing the prompt you used. For more complex projects, reporting the ways you used generative AI is more appropriate. For example, for this toolkit, generative AI tools were used to:

1. Brainstorm ways nurse educators may be able to use generative AI.
2. Classify ideas to create a cohesive and well-scaffolded framework for sharing the use cases.
3. Compare possible templates for communicating instructions for different use cases.
4. Synthesize our ideas about strategies to safely and efficiently use AI tools.
5. Create first drafts of instructions.
6. Provide first drafts of example prompts.
7. Generate suggestions for the document title.
8. Identify content gaps.

If being transparent about using generative AI is concerning, that may be an indication that it isn't the right tool for the task you are completing.

Is there a better tool for this task?

Generative AI can save time when used for tasks that involve structured pattern recognition or idea generation, such as brainstorming variations on a learning objective or rephrasing email text. However, it underperforms in tasks where the desired output is highly specific, especially when content must meet non-obvious criteria.

Sometimes, there is a tool designed for a specific use that will perform better than generative AI. For example, it is often faster to search literature using an indexed database than to search with a generative AI tool. Educators probably save more time using citation software compared to asking for generative AI to create citations and references. Relying on AI for tasks better suited to structured tools can reduce quality and increase workload.

Using AI as a search tool may not be as effective as using more specific search tools. For example, an educator searching for historic images to illustrate a lecture might consider using ChatGPT to generate links to potential images. But in this case, a direct Google Image Search with well-formed search terms would likely yield higher-quality results in less time. Asking AI to "generate a list of recent federal health policies affecting APRNs" may yield a mix of outdated laws, hallucinated legislation, or vague policy themes. Reviewing the latest updates from professional organizations and regulatory bodies may take longer at first, but yield accurate, citable, and professionally endorsed content. In these cases, nurse educators can use AI to help design search strategies to use with a more specific search tool.

Ultimately, educators determine if the generative AI is improving productivity or if it is shifting the cognitive work from the drafting phase to the editing phase. Educators may reduce their productivity if they spend large amounts of time assessing whether the AI's output is appropriate, whether facts are correct, and whether ideas are pedagogically sound. This layer of post-processing can add stress, particularly when the content is unfamiliar, high stakes, or subject to external review.

Does this Use Violate Privacy Laws?

Before uploading any documents to generative AI, consider the security provided by generative AI. FERPA, HIPAA, and institutional data governance policies may prohibit entering protected information into third-party AI tools. Before using AI to assist in grading or organizing clinical materials verify whether your AI use complies with institutional policies.

In general, educators can assume that any information they provide to the AI will be stored and used for model training and refinement. This means uploading student work for AI-assisted grading can violate FERPA unless the college or university has verified the tool meets FERPA requirements. Educators and students may violate HIPAA if they enter any patient identifiers into an AI tool that isn't explicitly covered by a business associate agreement. For example, a clinical instructor may upload case notes for AI to reorganize. If those notes include admission dates, the instructor has shared patient identifiers.

By treating all potentially sensitive data as strictly confidential and verifying compliance beforehand, educators can prevent unintended breaches and maintain the trust and privacy essential to their professional roles.

The Anatomy of Prompting

Educators are experts at sharing information and designing questions. Whether crafting exam questions, designing simulation debriefs, writing course objectives, or scaffolding class discussions, nurse educators are already skilled in the cognitive work that prompting requires. When educators design assignments, they think carefully about the audience, the purpose, and the level of instruction. They learn to ask questions that are neither too vague nor too narrow, and they tailor their language based on the developmental level of their students. These instincts translate into successful use of generative AI.

At its core, effective prompting depends on the same competencies: understanding context, anticipating how a question will be interpreted, and shaping responses to align with intended outcomes. Educators can adapt what they already do well to write effective prompts.

Why Prompting Matters

The quality of output from generative AI tools is entirely shaped by the prompt. Because these tools are probabilistic, not intelligent, they rely on how the user frames the task to generated different kinds of output. A vague prompt leads to vague results. A specific prompt generates more targeted, usable responses.

Consider the difference between these two prompts:

"*Write an essay on ethics.*"

"*Act as a nurse educator preparing a five-paragraph essay for recently graduated nurses on clinical ethics in neonatal care. Include three ethical dilemmas and relevant ANA guidance.*"

The first prompt is too broad and under-specified. The second gives clear guidance about audience, format, scope, and disciplinary context. This guidance will help the AI return a result that is usable with minimal revision.

Good prompting is the foundation of responsible AI use. It is how educators shape AI-generated content that reflects professional standards, reduces errors and hallucinations, and aligns with teaching or scholarly goals.

Common Prompting Mistakes

Educators new to AI often experience frustration because the results feel generic, shallow, or slightly "off." The issue is generally with the prompt. Common prompting pitfalls include:

- Being too vague: Prompts like "summarize this" or "make this better" give no guidance on purpose or audience.
- Overloading one prompt: Asking for too much at once ("write a case, generate references, explain implications") can lead to jumbled output.
- Lack of role, structure, or scope: If the AI doesn't know who it's supposed to be, how long the response should be, or what format is expected, it will guess.
- No audience targeting: Asking for "explanations" without indicating whether the audience is a policymaker, student, or peer can result in tone mismatches.
- Failing to fact-check: Even good prompts can return hallucinated or outdated content if the user doesn't review the result critically.

These challenges are invitations to apply the precision educators use in their instructional design work to prompt design.

The RISEN Framework

To support high-quality prompting, we recommend using the RISEN Framework developed by Kyle Balmer[10]:

R – Role: Who should the AI act as? Examples include a peer reviewer, a public health nurse, a curriculum designer, or a student.

I – Input: What source material, examples, or context should the AI use? Input refers to all the information you give the AI to start with, as attached materials, linked websites, and text in the prompt.

S – Steps: This is the instructions to complete the task. You will receive better output if you provide clear and structured instructions. For example, you can direct the AI to complete the task in a specific order.

[10] Manghani, S. (February 24, 2024). Prompt Engineering, Explained. Medium. Available at https://medium.com/electronic-life/prompt-engineering-explained-3b83ba347722

E – Expectation: This is the goal of the prompt, what you want it to produce. You can clarify how the output should be formatted (e.g., a table, an outline, a dialogue) and who the audience will be.

N – Narrowing: This is the limits you provide the AI. For example, you can ask it to draw from scholarly sources, limit the output to a single paragraph, or to follow a specific reporting style.

Example

Weak prompt: "Explain medication adherence."

RISEN-aligned prompt: "*Act as a nurse educator. Write a 150-word explanation of barriers to medication adherence in older adults for a student discussion board. Use plain language, avoid clinical jargon, and include one practical example*."

Breakdown of the RISEN Elements in the Example Prompt:

Role (R): "Act as a nurse educator" tells the AI to adopt a professional voice and perspective aligned with nursing education.

Input (I): While no specific document is provided, the phrase "barriers to medication adherence in older adults" sets a clear topic to work from.

Steps (S): " *Write a 150-word explanation*" tells the AI what task should be accomplished.

Expectation (E): The prompt asks for an "explanation" and specifies use in a "student discussion board," implying a conversational tone and educational framing.

Narrowing (N): "Use plain language, avoid clinical jargon, and include one practical example" provides tone guidance and audience targeting to ensure accessibility.

Prompt Refinement in Action

One of the most practical skills in AI use is learning to revise and layer prompts. Prompting improves through iteration, starting simple, refining based on the output, and being willing to adjust instructions.

Below is a prompt evolution that shows how a weak prompt becomes a strong, RISEN-aligned one.

Vague

"Write a teaching case."

Problems: No audience, scope, or structure. AI will guess

Slightly Better

“Write a case about gestational diabetes for nursing students."

Problems: Adds context, but still vague on format, learning level, or focus

RISEN-Aligned

"Act as a nurse educator. Write a one-paragraph teaching case on gestational diabetes for senior BSN students. Include one decision point and avoid medical management details."

Prompting in Phases

Rather than trying to get everything in one response, educators may find better results by breaking prompts into phases. This staged approach allows for more control and higher-quality outcomes. Each step builds on the last, keeping the model focused and reducing the risk of hallucination or drift. Prompting in phases mirrors lesson planning: start broad, refine with intent, build quality through iteration.

Example sequence:

"You are an expert nurse educator. List five key topics for a lecture on perinatal health equity for undergraduate students. The topics should align with the AACN essentials."

"Now write one learning objective for each topic using Bloom's taxonomy level comprehend."

"Organize the learning objectives into a lesson outline and provide a rationale for the scaffolding."

Closing Reflection

Prompting is a form of instructional dialog. Educators already possess the ability to frame clear, goal-directed questions. With modest practice and the RISEN framework as a guide, they can shape AI tools into effective collaborators. Strong prompts yield better results, reduce editing time, prevent frustration, and reinforce ethical, discipline-aligned use of AI in higher education.

Part Two: Course Design with Generative AI

Organizing Course Topics

Nurse educators may find generative AI particularly helpful when organizing course topics because it streamlines the complex process of aligning content with the AACN Essentials and broader program objectives. Generative AI tools can rapidly synthesize standards, identify redundancies, and highlight gaps, ensuring each course module contributes meaningfully to competency development. This saves time, promotes curriculum coherence, and supports accreditation-readiness by communicating the alignment between individual assignments and overarching educational outcomes.

AI supports creative flexibility by allowing educators to test and compare multiple approaches before committing to a specific course outline. Educators can ask AI to generate outlines for different teaching models then evaluate which approach best fits their topic, student needs, and program goals. In essence, AI acts as a versatile teaching partner, offering structured guidance without restricting academic freedom.

Overview of Process

Gather essential information

Gather your key information including your course description, credit hours, course format, a copy of the current syllabus or other source of suggested course topics, your program-level outcomes or competencies, your course-level outcomes or competencies, the AACN Essentials you are expected to meet, and any known constraints caused by university or department policies.

Upload Materials

Share the materials with the AI, either as an attachment or by including it in your prompt. Tell the AI that you are sharing these materials so you can begin planning a course and ask the AI if it can access all the materials. Provide anything it cannot read in a different format.

Identify Gaps and Redundancies

Use AI to evaluate your materials for completeness before you begin planning the course layout. For example, ask the AI to review your list of course topics and identify any gaps based on your courses expected learning outcomes.

Explore Sequence Options

Once you have a complete list of course topics, prompt the AI to suggest a few ways to sequence the course topics. You can ask the AI to evaluate each possible sequence for appropriate scaffolding, redundancy of topics, or ease of implementing specific teaching strategies or activities.

Finalize the course outline

You can ask the AI to perform a final evaluation of the "best" sequence to identify any potential issues and suggest ways to address the problems. Prompt the AI to prepare a final version of the course sequence that lists the topics, AACN Essentials, and course outcomes that will be addressed each week.

Prompt Examples

Review course topic list:

I'm planning a 10-week hybrid graduate course called Advanced Pharmacology for Nurse Practitioners. I'm including the course description, a draft topic list, and our program outcomes. Review these materials and identify gaps in the draft topic list before I start planning the full course sequence.

Ensure alignment of course topics with AACN Essentials:

Here is my list of course topics and the AACN essentials I will cover in this course. Align each topic with an AACN Essential, and determine if I need to add any topics to meet all the AACN Essentials I've been assigned.

Identify repeated concepts:

Do any of the proposed weekly topics overlap or repeat in a way that might confuse students or reduce the effectiveness of scaffolding. If you find redundancies, recommend ways to consolidate or differentiate them.

Develop course sequence:

Create three different course topic sequences using this list of topics. Make one sequence emphasize concept scaffolding, one that prioritizes clinical relevance, and one that clusters content around major assessments.

Evaluate possible course sequences:

Compare the three possible course sequences you just generated in terms of how well they support scaffolding, teaching strategy flexibility [for a flipped classroom], and avoidance of redundancy.

Summarize course plan:

Create a final weekly layout for this course that includes: 1) the topic, 2) the AACN Essentials aligned with that week, and 3) the course-level outcomes addressed. Format it as a table so I can use it for curriculum mapping.

Next Steps

Generative AI can help you backward design a course. To achieve this, start with your final assignments and the required course outcomes. Ask the AI to suggest a weekly topic sequence and teaching strategies to ensure the students are fully prepared for each assignment.

AI can suggest opportunities to integrate program-level requirements into an existing course plan. For example, if your program requires inclusion of social determinants of health in each course, you can provide the AI with your course plan and ask it to suggest where it could be integrated.

AI can help you plan for the unexpected, such as having to cancel a class meeting due to weather. To do this, give AI your course plan and ask it to predict the impact of missing a week. You can also ask the AI to suggest ways to adjust the course if you miss a week.

If your course includes multiple levels of learners, such as pre-licensure students and RN-to-BSN students, you can ask the AI to suggest ways to adapt the weekly topics for both learner groups without changing the core sequence.

Generative AI can map your program alignment to the AACN Essentials or your program outcomes by evaluating course outlines for multiple courses in your program. This is a great way to identify gaps and determine whether redundancies are intentional scaffolding.

Writing Module Objectives

Educators can face challenges crafting learning objectives that strike the right balance between specificity, cognitive complexity, and alignment with professional standards. Generative AI tools like ChatGPT can serve as intelligent writing partners, helping educators brainstorm, refine, and align module objectives quickly and efficiently. By prompting the AI with key information (such as the topic, target learner level, and relevant standards) educators can generate draft objectives that are tailored to their instructional needs. Using AI in this way supports instructional coherence.

Educators can use generative AI to scaffold objectives across modules, check for measurable action verbs aligned with Bloom's Taxonomy, and ensure consistency with course-level outcomes and regulatory frameworks. Whether developing new content or revising existing modules, AI can provide multiple versions of an objective, offer suggestions to increase cognitive rigor, and even highlight alignment with clinical competencies. This collaborative process empowers educators to focus more on the pedagogy behind their teaching and less on the logistics of objective formulation.

Overview of Process

Clarify module topic

Define the scope of your module by clarifying the topic and the purpose of the module. Define your audience level, which may be as simple as master's level students or as specific as prelicensure students in their second semester of the program.

Draft module objectives

Identify the anchors for the objectives, which may be course-level objectives, the AACN Essentials, or clinical competencies the module supports. Choose the cognitive level for learning. For example, are the students expected to apply the information to new contexts or evaluate arguments based on the information? Use this information to prompt the AI to generate a list of possible objectives.

Review and Refine

Review the output and assess the AI generated objectives. If necessary, ask the AI to make revisions. Select the objectives that best match your teaching

context and edit for clarity and language. Check the selected objectives for alignment with course objectives and logical progression from prior modules.

Prompt Examples

Describe the Module and specify learning level and outcomes

I am creating a module for a graduate nursing course. The topic is 'de-implementation of ineffective practices in clinical settings.' The students are in a Doctor of Nursing Practice program, and this module falls in Week 12 of a systems improvement course. I want the module objectives to use Bloom's Taxonomy verbs at the 'analyze' and 'evaluate' level. Students will be expected to assess practice patterns and recommend changes.

Request the initial objective drafts

Generate 4 module-level learning objectives using this list of module topics and these AACN Essentials. Make sure they are measurable and appropriate for DNP-level learners

Add alignment to professional standards

Revise or align these objectives with AACN Essentials 7.1h (Uses system knowledge to make recommendations for resource allocation and policy development) and 5.1l (Evaluates organizational systems that affect care delivery and patient outcomes)

Improve Clarity and Accessibility

Simplify the language of these objectives to improve readbility and ensure each objective can be assessed through written discussion, case analysis, or project work.

Check for Progression Across Modules

Here are the objectives from last week's module, which focused on stakeholder engagement in quality improvement. Do these new objectives build logically on the previous week's goals?"

Next Steps

Ask the AI to suggest assignments, discussions, or simulations that directly assess each module objective.

Use the AI to help sequence learning by building cognitive complexity across weeks.

You can use generative AI to help you build a curriculum map that aligns the module objectives with the course objectives and frameworks like the AACN Essential or QSEN competencies.

Use the AI to analyze your list of existing module objectives and highlight potential gaps in competency domains, cognitive levels, or skill-based objectives.

Preparing Lesson Plans

AI tools can help you brainstorm engaging activities tailored to specific learning objectives and help you explore innovative teaching methods. The ability to generate a variety of activity suggestions and formats supports the incorporation of new approaches, helping to foster a dynamic and stimulating learning environment.

Collaborating with generative AI will also support the preparation of course materials that are cohesive and well-aligned with the lesson plan. In this way, AI can help you streamline the administrative burden that often accompanies lesson planning so you can focus on pedagogical innovation and personalized student engagement.

Overview of Process

Provide the AI with your planning information

Begin by providing the AI your courses' learning objectives and desired outcomes. Be specific about the objectives and outcomes for the lesson you are planning and how it fits into the overall course objective.

Brainstorm activity Ideas

Ask the AI to generate a list of activities that are appropriate for your topic, student level, and timeframe. Ask the AI to evaluate each of the suggested activities for alignment with the lesson or course objectives. You can ask the AI to make the assessments based on best practices. Use the AI's evaluation to help you select the activities you would like to complete.

Prepare course materials

Identify the activities you want in the lesson plan and ask the AI to prepare course materials for those activities. This may be student handouts, a lecture outline, or activity directions.

Evaluate the final plan

Ask the AI to evaluate the final lesson plan to identify gaps, redundancies, alignment with objectives, or logistical issues. For example, you can ask the AI to determine if you will have enough time to complete the activities, if the activities are appropriate for the level of learner, or if additional preparation will be needed for students to complete the lesson based on course sequence.

Prompt Examples

Define Learning Objectives and Outcomes:

Help me define clear, measurable learning objectives for a lesson on [specific topic], focusing on [specific skills or knowledge]. How can these objectives fit into the broader course goals of [course name]?

Generate activity ideas:

Generate a list of engaging, interactive activities suitable for [topic] that align with the learning objectives. The lesson duration is [X] minutes, and my students are at the [undergraduate/graduate] level. Include a mix of activities such as group discussions, case studies, role plays, or individual assignments.

Evaluate activity alignment:

Evaluate the following activities for their alignment with the learning objectives I've defined: [list activities]. Based on best practices in teaching [discipline], suggest which activities are most effective in meeting these objectives and provide the rationale for your decisions.

Generate additional materials:

For the selected activities [list chosen activities], help me create a lecture outline with key points and timing for each section.

For the selected activities [list chosen activities], help me create student handouts that summarize key concepts and provide instructions for the activities.

For the selected activities [list chosen activities], help me create directions for group work or individual tasks that clearly outline the expectations and goals.

Evaluate the lesson plan for feasibility

Review the completed lesson plan with the following activities: [list activities]. Will the time allocated for each activity be sufficient for students to engage meaningfully?

Are these activities appropriate for master's level nurse students? [list activities]

Finalize the plan:

Given the overall course sequence, check if this lesson plan aligns with the upcoming lessons. Are there any potential challenges or adjustments I can consider so the lesson flows smoothly into the next session?

Next Steps

Generative AI can help you continuously improve your lesson plan based on student feedback. Provide the AI with your lesson plan and the student feedback and ask for suggestions to improve the lesson.

Ask the AI to include real-time formative feedback into the lesson plan during the activities. The AI can suggest tools or methods that will be easily integrated into the lesson.

Ask the AI to identify ways to incorporate inter-disciplinary activities into the lesson. AI can suggest disciplines to include and ways to include them.

Generative AI can be used to create lesson plans around specific activities, such as case study, role-play, or games. You can use AI to create the activity and use the AI to reverse-engineer the lesson around the activity to ensure students are fully prepared.

Ask the AI to draft a test blueprint based on lesson plans across multiple days of class.

Preparing Lecture Outlines

AI tools can help educators break down complex topics into structured, understandable lecture sections and automatically generate an outline that aligns with specific learning objectives. The outline can include the key concepts, subtopics and suggested teaching strategies to ensure the lecture flows logically and maintains student engagement.

AI can also help you evaluate your existing lecture outlines, identifying gaps and redundancies. Educators can prompt AI to suggest instructional strategies that align with the lecture topics and design activities for formative assessment during class. This collaboration allows educators to refine their lecture approach and focus more time on preparing engaging content and anticipating student questions, leading to a more impactful and dynamic student experience.

Overview of Process

Gather the planning information

Include information about the level of student attending the lecture, the overall course topic, and the program the student is enrolled in. Define the learning objectives and student outcomes for the lecture. Identify the key concepts and subtopics that you want to cover in the lecture. If these have not already been defined, you can prompt the AI to help you generate a list of concepts and subtopics.

Explore potential sequences

Prompt the AI to structure the lecture flow by organizing the key concepts and subtopics into a logical sequence. Ask the AI to provide multiple versions and compare them for learning effectiveness.

Explore interactive strategies

Prompt the AI to suggest interactive strategies and activities to engage the students during the lecture. Using iterative prompting, ask the AI to evaluate the strategies based on relevant constraints, such as the lecture format (online vs in person), size of the audience, background knowledge, or available time. Work with the AI to integrate your selections into the lecture outline.

Evaluate and refine

Use the AI to help you evaluate the outline for coherence and alignment with the learning objectives. You can also ask it to evaluate the time it will take for each section. Ask AI to make suggestions for any issues it identifies.

Finalize your outline

Adjust and finalize the outline based on the AI feedback. AI can do this step, but you will need to make the final decision to ensure the content reflects your teaching style.

Prompt Examples

Draft learning objectives

I am preparing a lecture on [specific topic] for my [undergraduate/graduate] students in [course name]. The goal of this lecture is for students to understand [specific learning objectives], including key concepts such as [list of subtopics]. I would like the lecture to last approximately [X] minutes, with a mix of lecture content, discussion, and interactive activities. Please suggest ten measurable learning objectives for this lecture based on the topic and key concepts provided.

Compare objectives

These are the five lecture objectives I am considering. Review the objectives and identify any areas of overlap or potential content gaps for a lesson about [specific topic].

Lecture outline

Using this list of learning objectives, the topics, and key concepts, create a detailed lecture outline with the following sections:

Introduction (approx. [X] minutes): Key points to introduce the topic

Core Content (approx. [X] minutes): Breakdown of key concepts/subtopics

Interactive Activity (approx. [X] minutes): Suggestions for student engagement, such as case studies, group discussions, or Q&A

Conclusion (approx. [X] minutes): Summarize the key takeaways and set up for the next session

Integrate teaching strategies

Suggest interactive strategies or activities that would encourage student engagement and reinforce the learning objectives during this lecture. These could include multimedia, real-life examples, or group work.

Evaluate content

Assess whether the outline is balanced and aligned with the learning objectives. Are there any gaps or redundancies? Suggest ways to address any issues you find.

Suggest supporting materials

Based on the outline, generate recommendations for materials such as lecture slides, handouts, or additional readings that would reinforce the lesson content.

Evaluate the lecture outline

Review the outline and activities and asses if they are engaging and appropriate for students at the [undergraduate/graduate] level. Then provide time estimates for each lecture section."

Next Steps

Use student performance trends to personalize the content for each class. Provide the AI with information about the course topics that have been the most challenging for students in this class. Ask the AI to suggest simple changes to the lecture outline to help reinforce learning on difficult course topics.

Prompt the AI to generate a student self-assessment tool so your students can evaluate their understanding of the lecture.

Ask the AI to generate practice questions or NCLEX-style questions from your lecture outline.

Allow AI to create a lecture outline from your prior lecture recordings. Use this outline with the AI to evaluate and refine your lecture plan.

Prompt the AI to prepare a lecture guide or study guide for students based on the lecture outline.

Use AI to turn old lecture outlines into Video Scripts for an online course. AI can suggest sections for shorter videos rather than one long video lecture.

Test Blueprint

Rather than replacing educator expertise, AI acts as a creative and analytical partner that helps educators move more quickly from ideas to draft test blueprints. AI can synthesize course objectives into an initial plan, suggest ways to distribute questions across content areas and cognitive levels, and identify possible gaps or misalignments with program standards. Educators can then focus on refining the blueprint instead of building it entirely from scratch. By accelerating the more mechanical aspects of blueprint creation, AI frees up educator time for higher-level thinking and ensures that assessments remain both rigorous and learner centered.

Overview of Process

Gather essential materials

Start by collecting your course syllabus, learning objectives and outcomes, accreditation standards (such as AACN Essentials or specialty guidelines), and any existing course outlines or previous blueprints. You'll also want to include skills checklists, clinical competencies, and any program-level requirements that influence content priorities. AI can assist even at this early stage by helping summarize and organize documents. You can upload your syllabus and objectives to an AI tool and ask it to extract key outcomes, which saves time and ensures you don't overlook critical content areas.

Organize objectives and prioritize content

Once all objectives are listed, sort and rank them based on importance, frequency in the course, and their relevance to safe, effective practice. Decide which objectives will be heavily emphasized and which might receive lighter coverage or no questions. AI can help analyze and categorize objectives, suggesting groupings (e.g., by body system, topic, or skill domain) or even proposing an initial prioritization based on keywords or course emphasis. This can provide a helpful starting point that you can review and adjust.

Confirm appropriate cognitive levels

Assign each objective a cognitive level to demonstrate mastery (e.g., knowledge, application, analysis). AI can assist by suggesting cognitive levels for each objective, flagging which are likely to involve recall versus application or

analysis. This is especially useful if you have a long list of outcomes, as AI can provide a first-pass classification for you to verify and refine.

Draft an initial blueprint prompt for AI

Combine your prioritized objectives and cognitive levels into a clear prompt for the AI. Include the total number of test items and specify your distribution preferences (e.g., more application-level questions, minimal recall items). AI shines in this step by generating an initial draft blueprint that shows how many questions to assign per objective and cognitive level. You might say, "Using these objectives and Bloom's levels, create a test blueprint for a 75-question exam emphasizing application and analysis." The AI can quickly produce a detailed table or matrix as a starting point.

Review and evaluate AI-generated blueprint

Carefully examine the AI-generated blueprint to check if it aligns with your intended focus areas and program standards. Look for any imbalances, underrepresented objectives, or unexpected emphasis. AI can also help here by providing alternative versions on request, such as one with more emphasis on application skills or one balancing all cognitive levels equally. You can ask it to highlight differences between versions or generate summary insights about coverage gaps to make your review more efficient.

Iterate and refine

AI makes this step much faster by rapidly regenerating updated versions each time you adjust your instructions. Provide specific feedback to the AI or refine the prompt to clarify any misalignments you noticed. Adjust the blueprint until it fully matches your goals. AI can help you compare versions side by side and suggest ways to fine-tune item distributions.

Validate with colleagues or standards

Once refined, validate the blueprint against course standards, program expectations, and accreditation requirements to ensure it is comprehensive, fair, and aligned. AI can help summarize your blueprint or prepare visual representations (like charts or heatmaps) to make it easier for others to review and provide feedback. This visualization can facilitate more productive educator discussions.

Finalize and document

Make final adjustments and document your rationale for content emphasis and cognitive level choices. Save and share the final blueprint for future use, accreditation reports, and course archives. AI can generate a polished, formatted version of the blueprint, including summary tables and a narrative explanation of your decisions. This can make your documentation more consistent and easier to share, supporting transparency and continuous quality improvement.

Prompt Examples

Organize objectives and prioritize content

Here is a list of my lecture topics, course objectives, and learning outcomes. Please group the lecture topics into major themes or content areas and suggest a priority level for each topic (high, medium, or low), based on their importance to overall course goals and safe nursing practice.

Decide on cognitive levels

Using Bloom's Taxonomy, assign a cognitive level (knowledge, comprehension, application, analysis, synthesis, or evaluation) to each of these objectives. Explain why you selected each level so I can review and adjust as needed.

Draft an initial blueprint

Using these prioritized objectives and assigned Bloom's levels, draft a test blueprint for a 75-question exam. Suggest the number of questions per objective and distribution of cognitive levels for objective questions. Emphasize application and analysis skills and ensure that high-priority objectives have more questions than medium or low-priority ones.

Review and evaluate AI-generated blueprint

Analyze this draft blueprint and identify any potential gaps or areas that may be over- or underrepresented. Suggest adjustments to improve balance and ensure alignment with my priorities and Bloom's level distribution.

Iterate and refine

Based on my feedback (e.g., 'Objective 3 needs more questions,' 'too many knowledge-level items'), revise the blueprint to adjust item distribution. Provide

an updated version and summarize the changes you made compared to the previous version.

Next Steps

Use AI to generate draft test questions based on the blueprint (e.g., multiple-choice, case-based, short answer), which educators can then review and refine.

Develop alternative versions of exams (parallel forms) to support test security and accommodate makeup exams, with consistent content coverage and cognitive level balance.

Build study guides or exam preparation outlines for students, directly aligned with the test blueprint, to enhance transparency and support focused studying.

Design formative quizzes or self-assessments based on blueprint objectives, giving students additional low-stakes practice opportunities.

Map course-level blueprints to program-level outcomes and accreditation standards to ensure curricular coherence and identify gaps across courses.

Develop competency checklists or skill validation tools that mirror cognitive levels and content emphasis from the blueprint, supporting clinical assessment.

Draft narrative summaries explaining blueprint choices and assessment design for accreditation reports, curriculum committees, or external reviewers.

Use AI to simulate different blueprint scenarios (e.g., increasing higher-order questions, reducing recall) to understand potential impacts on student performance.

Part Three: Assignment Design with Generative AI

Discussion Questions

Collaborating with AI to generate discussion questions can quickly generate a wide range of thought-provoking questions that encourage critical thinking, problem solving, or application of course concepts. Nurse educators can specify the subject matter, desired learning outcomes, and level of complexity. With that information, AI can draft relevant, high-quality prompts quickly. Educators can use these prompts to spark debate, reflect on case studies, or challenge students to consider different perspectives.

AI can also be used to edit existing discussion prompts to meet the needs of students at different learning levels, in different courses, or from different points of view. Generative AI can also help you create a large pool of discussion prompts, ensuring balance between prompts that ask for factual recall, application, and analysis.

Overview of Process

Prepare the background information

Begin by defining the learning objectives you plan to meet with the discussion, the student level, and the course context. Course context may include a lecture outline, a list of readings, or a case study. Provide this information to the AI with the prompt to help guide the idea generation.

Consider the type of discussion

Specify the type of discussion question you want the AI to draft. These may be open-ended questions, a problem-solving scenario, issues to debate, a rhetorical question, or something else. Consider how complex the question will be, the discussion format, and amount of time for discussion. The AI can use this information to make suggestions about the types of questions that best fit the learning objectives and student level.

Draft and refine questions

Prompt the AI to generate a variety of prompts based on the information you prepared in steps one and two. During this step, you can prompt the AI to generate questions for different blooms levels, from different perspectives, or on the same topic but for different courses. The AI can help evaluate the questions. If you ask the AI to review questions for clarity, relevance to the lesson objectives

and course content, and appropriateness for the student level. You can ask AI to make refinements to the list based on this evaluation.

Create your active learning strategy

Once you have your discussion question, AI can help you brainstorm and evaluate strategies for how to use the discussion questions to generate engagement. Generative AI can also suggest adjustments to the questions so they are more likely to stimulate engagement.

Prompt Examples

Generate ideas based on objectives

We are going to generate a list of discussion topics for a lecture on health policy. The learning objectives are to understand key health policy concepts, analyze their impact on different populations, and evaluate the role of nurses in advocacy for access to healthcare. Suggest possible topics for the discussion prompts and how each prompt aligns with the learning objectives

Specify the type of discussion prompt

Create a set of discussion questions for undergraduate students in their first health policy course. Include open-ended questions, a case study analysis, and problem-solving prompts that require students to critically analyze how health policy affects the work of nurses. The students have prior knowledge of hospital care and are new to basic policy concepts.

Evaluate and Refine the Prompts

Evaluate the following discussion questions for their relevance to health policy and social justice. Are these questions aligned with the learning objectives, appropriately challenging for graduate students, and diverse in their focus?

Adjust for Interaction and Engagement

Revise these discussion questions to encourage peer-to-peer interaction. Include prompts where students can respond to or build on each other's answers.

Next Steps

Provide the AI with information about student performance on specific course objectives or topics to generate discussion questions tailored to students' current

understanding and learning needs. AI can help analyze areas of difficulty or gaps in knowledge to create discussion questions that will reinforce learning in these areas.

Ask the AI to generate discussion questions for a simulation or role-play that will require students to draw connections between the different perspectives.

Ask the AI to generate discussion questions that will bridge major concepts between courses, such as asking students to draw on learning from a prior class on health policy as part of a discussion on nursing leadership.

Use AI to generate discussion questions based on real-world current events, trends, or emerging issues in the field. The AI can link the real-world example to the course content. For example, you could ask the AI to generate discussion questions for different courses based on a current event, allowing educators to generate rich dialog about the role of nursing within the broader context of the community.

Use AI to create a discussion facilitation guide that provides information about how to introduce the discussion, what topics to cover before the discussion, and how to transition from the discussion to the next activity.

Assignment Deconstruction

Because generative AI can produce good writing, educators may need to redesign large writing or project assignments that do not include draft submissions. By using AI to break down large assignments into smaller, more manageable components, educators can increase student engagement with the course work while also making it more difficult to rely solely on AI-generated content. This strategy allows educators to track student progress and provide formative feedback at the early stages of the assignment. Strategies like integrating peer reviews, requiring multiple sources, and requiring students to submit work at multiple stages encourage deeper reflection and analysis. These adjustments help ensure students are actively engaged with the material and demonstrating understanding.

Educators can harness the creativity of generative AI to examine multiple approaches to assignment deconstruction, align the smaller assignment parts with the course outline, and evaluate the series of assignments for meeting the course objectives. In this way, educators can incorporate a variety of strategies to encourage students to do the "thinking" on their own.

Overview of Process

Clearly define the student learning outcomes

Begin by revisiting the learning outcomes or objectives of the assignment. Identify the skills and knowledge that students will gain from completing it. Where possible, modify the objectives to emphasize skills like analysis, application, and synthesis of knowledge because AI cannot easily replace the process of critical thinking. You can use AI to help you rewrite the assignment objectives if necessary.

Design assignment segments

Prompt the AI to break the project or paper into a set of preliminary assignments for the student to complete and submit. Ideally, these will be based on the work the student will need to complete for the full project. Ask the AI to give you multiple versions so you can compare and select the best options.

Select your assignment groups

Review all the suggested assignment segments and make your preliminary selections for a group of assignments. Use the AI to evaluate the alignment of

the assignments with the assignment objectives and the overall course objectives. Ask the AI to provide suggestions for how to address any gaps.

Sequence the assignments

Use the AI to explore the sequencing of the assignments given the course outline. Identify any potential conflicts where students will not have enough time to complete the assignments. Finalize your assignment selections and work with the AI to prepare student instructions and other assignment materials.

Prompt Examples

Revisiting Learning Outcomes or Objectives

Help me review and refine the learning objectives for this assignment on [topic]. The goal is to ensure the objectives emphasize critical thinking, analysis, and application of knowledge. Rewrite the objectives to better reflect skills like synthesis and evaluation, rather than just fact recall. Here are the current objectives:

[list of current objectives]

Generating Alternative Assignments

Break down this assignment on [topic] into three to five distinct steps for students to complete. Generate an assignment for each step that would meet the learning objectives of [assignment topic] while being difficult for students to complete using generative AI. These assignments should require original thinking, synthesis of multiple sources, or application to real-world scenarios. Provide multiple versions so I can evaluate which approach best promotes deep engagement and critical thinking.

Evaluating Assignment Alignment with Objectives

Evaluate the following proposed assignments for alignment with the learning objectives of [assignment topic] and the overall course objectives. Here are the alternatives I'm considering: [list of alternative assignments]. Provide information about any gaps you identify and suggest ways to strengthen the assignments to better align with the course outcomes.

Exploring Sequencing of Assignments

Given the course outline and calendar, help me explore the sequencing of these assignments. Are there any potential conflicts where students may not

have enough time to complete the assignments? For each step of the assignment, suggest a timeline that considers course workload and ensures appropriate pacing without overwhelming the students."

Finalizing Assignment Selections and Preparing Materials

I selected this assignment on [topic]. Prepare the instructions and any additional materials needed for students. I need clear and concise instructions that guide students through each step of the process. Include suggestions for rubrics, criteria for evaluation, and any supporting resources that will help students succeed.

Next Steps

Use the strategies to deconstruct an assignment to update your assignments based on real-time data or current events. Prompt the AI to create assignments that require students to analyze the latest research, news reports, or policy changes. This keeps your assignments fresh and applicable, engaging students in real-world problem solving.

Build assignments around specific readings, requiring the students to reference multiple readings from the course into their submission.

Synthetic Data Generation

Co-creating synthetic data with generative AI can ensure the data have qualities needed to highlight specific concepts. Synthetic data refers to information that is artificially generated to resemble the structure and statistical patterns of real-world data, without exposing any actual patient records. Educators can use AI-generated synthetic data to build activities and assignments that challenge students to think critically. As with other information generated by AI, the data is provisional until the educator tests and confirms that the data meets the properties necessary for the learning activity.

Examples of Synthetic Data

Clinical Reasoning Through Dynamic Case Studies

Synthetic data allows nurse educators to develop rich, evolving clinical case studies that go beyond static narratives. These cases can include changing lab values, vital signs, progress notes, and medication administration records that unfold over simulated time allowing students to engage in repeated decision-making. For example, students can adjust nursing care plans and prioritize interventions based on changing information. This type of activity encourages deeper clinical reasoning and mimics the complexity of real-world patient care. Educators can generate multiple versions of a case to explore similar diagnoses, health disparities, or variations in treatment response without relying on actual patient data.

Documentation Practice

AI-generated synthetic data provides a flexible resource for helping students strengthen their clinical documentation skills in a structured and realistic way. Educators can provide students with generated patient charts that include vital signs, assessment notes, lab results, and medication records. Using these data, students can practice extracting key clinical information and writing concise, situation-specific SBAR (Situation, Background, Assessment, Recommendation) reports. In addition, educators can ask students to complete simulated documentation forms such as nursing progress notes, intake assessments, or discharge summaries. Using synthetic data for documentation practice reinforces both clinical judgment and professional communication standards while avoiding the ethical and legal constraints associated with real patient data.

Quality Improvement and Systems Thinking

Synthetic datasets can be tailored to mimic quality indicators such as fall rates, hospital-acquired infections, or readmissions. Nurse educators can use these datasets to help students learn to identify trends, conduct root cause analyses, and propose quality improvement (QI) initiatives. For example, students might use synthetic patient safety data to build fishbone diagrams, control charts, or Plan-Do-Study-Act (PDSA) cycles. This application of synthetic data supports systems thinking, encourages collaboration, and links classroom learning to the real-world challenges of improving patient outcomes in complex healthcare settings.

Overview of Process

Define the Educational Objective and Data Scope

Clarify what you want students to learn and what kind of data you need from the AI: individual patient cases, population health tables, nursing notes, lab results, or narrative reports.

Draft a Precise Prompt for the AI

Write a detailed prompt that tells the AI exactly what to generate. Include:

- Patient demographics (age, gender, relevant history)
- Clinical setting (e.g., ED, med-surg, community clinic)
- Target data types (e.g., vital signs, SOAP notes, discharge instructions)
- Level of complexity based on learner readiness

Review and Refine the Output

Carefully read the AI-generated output. Check for clinical plausibility, do the symptoms and labs match the diagnosis? Review the internal consistency, for example that the vital signs match the acuity level. Determine if the data are clear and have education value, that the data will support the intended lesson. If needed, revise your prompt to correct errors or add missing elements, and regenerate.

Format the Data for Student Use

Convert the raw output into a usable format. This is achieved by prompting the AI to create the specific output you need. For example, the AI can convert the data to a narrative case study, a table, or a simulated patient chart. You can also segment the output into multiple components to support stepwise learning

(e.g., reveal vitals only after students assess history). Depending on the AI tool you use, it may be able to provide you with a document ready for distribution or code ready to upload to your LMS.

Annotate with Teaching Notes

Add your own comments or teaching notes for student or grader use. You can highlight key learning points and identify red flags or decision-making moments. This transforms raw AI output into a guided learning activity. If needed, you can prompt the AI to provide debrief questions.

Prompt Examples

Synthetic Patient

Generate a synthetic patient case for a 65-year-old male with congestive heart failure admitted to a medical-surgical unit. Include a brief history, vital signs, home medications, reason for admission, and two nursing concerns.

Synthetic Documentation and SBAR

Provide a brief patient profile and clinical findings for a 79-year-old with pneumonia. Then generate an SBAR report that a nurse might give to the attending physician during the night shift.

Synthetic Quality Data

Generate synthetic data for a quality improvement case at a mid-sized urban hospital. The focus is on reducing patient falls on the medical-surgical unit. Provide four brief patient profiles representing fall incidents from the past month, including relevant demographics, time of fall, contributing factors, and nurse shift. Then include a table summarizing monthly fall trends over six months and a short nurse manager note summarizing staff concerns. The case should be realistic and appropriate for BSN students learning QI methods.

Synthetic Practice Data

Create a synthetic dataset showing rates of adolescent depression screening over one year at a school-based clinic. Include monthly counts, demographic breakdowns, and a short reflection from a school nurse.

Next Steps

Develop different versions of the data for various purposes such as practice sets for low-stakes skill building.

Introduce students to AI as a data generating tool and critical assessment of AI-generated content. Discuss the ethical use of synthetic data and the limitations and bias in generated content.

Use synthetic data to explicitly teach and assess elements of NCSBN Clinical Judgement Measurement Model. Scaffold students thinking by prompting them to recognize cues, analyze data, and prioritize responses in unfolding synthetic cases.

Co-Writing Case Study Narratives

Generative AI can quickly create complex, realistic scenarios, encouraging creative exploration of multiple perspectives and system actors. It helps maintain consistency in tone and structure, supports inclusive design by suggesting underrepresented viewpoints, and allows for rapid iteration to test different storylines or simulate "what if" scenarios. This flexibility enables educators to focus more on aligning content with learning objectives and refining critical details.

However, co-writing case study narratives with AI has important challenges. The stories may feel generic or lack nuanced depth without careful guidance. There is potential for inaccuracies or embedded biases that reflect limitations in AI training data. Unedited AI output can lack local contextual relevance which is essential for effective case studies. To overcome these drawbacks, educators provide strong oversight, purposefully craft prompts, and integrate their own clinical expertise and institutional knowledge.

Overview of Process

Define Learning Objectives

Define the goals of the case study and the specific skills the students will develop or demonstrate. This information will help to narrow the information and keep the case focused.

Gather contextual and source information

Compile any relevant clinical guidelines, policy documents, or organizational data needed for the case. Define your key actors in the case and their roles. AI can help you create roles by providing you with options for the scenario and key actors.

Create a high-level outline

Establish a framework for the case including the central problem and possible decision points. If the learning objective isn't about specific decisions, you can use AI to help you explore example problems and decisions that would allow students to demonstrate a skill.

Create a detailed AI prompt

Write a detailed AI prompt using the information about the objectives, context, setting, scenario, and key actors you created. Include information about the length of the narrative. If the narrative is long, you can prompt the AI to write shorter sections (a few paragraphs) at a time.

Review and Revise

Check the narrative for accuracy, clinical realism, and alignment with objectives. Remove any content that is too generic and add nuanced details. You can ask the AI to rewrite the narrative according to the edits you want made.

Prepare Supporting Material

You can ask AI to create supporting material for cases such as internal memos, root cause analysis reports, or patient telephone messages.

Example prompts

Initial Framework or Scenario Outline

Create three options for a case study involving a midwife caring for a client with gestational diabetes in a rural clinic. Include main characters, setting details, key challenges, and possible decision points.

Detailed AI Prompts for First Draft Narrative

Write a 600-word narrative describing a nurse practitioner introducing a new evidence-based asthma management protocol in a busy urban community health center. The learning outcome is to explain the steps of the PDSA cycle. Include staff reactions, patient concerns, and at least one unexpected complication that requires adaptive leadership.

Refinement and Deepening Prompts

Adjust this case study narrative to reflect policies and common practices in Nevada hospitals, including specific references to local workforce shortages and cultural diversity among patients.

Supporting Materials Prompts

Create a mock meeting summary that would accompany this case study narrative for students to analyze as an artifact. The meeting summary will provide the results of the root cause analysis.

Final Check Prompt for Educator

Identify any areas in this case narrative that may need additional detail or revision to ensure accuracy, realism, and alignment with advanced nursing practice standards.

Next Steps

Use the AI to draft classroom discussion questions or discussion board prompts based on the case study.

Ask the AI to generate multiple versions of the case study so the students can compare the factors that resulted in different outcomes.

Generate a case study that builds each week based on the content of the course.

Evaluating Cognitive Complexity

Educators frequently analyze learning objectives, assignments, and assessments to determine their level of cognitive complexity and ensure alignment with desired course outcomes. Generative AI can support this process by quickly analyzing the language used in instructional materials (e.g., verbs, question stems, and task requirements) and classifying them based on standardized systems such as Bloom's Taxonomy. Generative AI can also help identify inconsistencies, redundancies, or gaps in cognitive rigor and suggest enhancements to better match course goals and accreditation standards. Educators who complete this work across courses can synthesize the results into a curriculum evaluation.

Overview of Process

Determine the Purpose

Determine whether the goal is to assess existing learning objectives, rewrite objectives to meet a higher or lower level, or evaluate alignment between objectives and assessments. This will help you write an effective prompt.

Provide Content for Review

Provide the AI with course or module objectives, assessment questions, or assignment descriptions for evaluation. This can be done by typing in the prompt box or by uploading documents and then prompting AI with the direction of how to analyze the attached documents.

Use Clear and Specific Prompts

Clearly describe what you need. For example, ask for classification of each objective by Bloom's level. Alternatively, you may prompt the AI to make suggestions to elevate cognitive complexity. Ask for downloadable or editable versions as needed. The AI will analyze the verbs and structure of objectives or assessment items and classify them into the appropriate Bloom's taxonomy classification. It can also provide explanations and offer higher-level alternatives if needed.

Review and Edit

Review AI-generated classifications and edit as needed based on your course goals or assignment standards. Adjust any misaligned or unclear objectives.

Cross-check Bloom's levels against course outcomes and ensure consistent alignment across lectures, activities, and evaluations.

Prompt Examples

BSN: NCLEX-Style Question Review

Review the following NCLEX-style multiple-choice questions intended for BSN nursing students. Classify each question according to Bloom's Taxonomy. If any questions are below the 'Apply' level, revise them to reflect more clinical reasoning or decision-making. Present the results in a table that includes the original question, Bloom's level, rationale, and revised version.

MSN: Literature Review Assignment

Analyze this literature review assignment for an MSN research methods course. Identify the cognitive demands based on Bloom's levels. Recommend edits to the prompt to elevate it from 'Understand' and 'Analyze' to 'Evaluate' or 'Create', where appropriate. Provide a side-by-side comparison of the original and revised versions.

Exam Blueprint Audit

Analyze this exam blueprint for a pharmacology course. Classify each question type or topic according to Bloom's level. Identify gaps in cognitive variety and recommend revisions to the blueprint to ensure alignment with course objectives and expectations for graduate-level application and synthesis.

Next Steps

Use the results from your assessment of all learning activities to create a cognitive complexity map for the course.

Use generative AI to help draft parallel questions (questions on the same topic at the same cognitive complexity) to build a bigger test question pool.

Use the assignment complexity information to help design a rubric.

Rubric Development

Rubric development involves identifying clear criteria for evaluating an assignment or performance and defining levels of achievement for each criterion. Generative AI can help ensure consistency, reduce bias, and streamline the rubric creation process, making assessment more efficient and data-informed for educators and evaluators. Alternatively, AI can suggest revisions to a rubric to align the rubric with course outcome criteria and descriptive indicators in the assignment. AI can also help educators identify mismatches between assignment instructions and grading rubrics.

Overview of Process

Input Materials

Specify the learning goals, assignment type, or skills to be assessed. This may include module, course, or program goals. Upload relevant documents to the generative AI. This will include the assignment description and any existing rubrics. If you have example assignments, you can upload those as well.

Input Detailed Prompting

Prompt the AI to use the materials to evaluate or develop the rubric. If you already have the evaluation criteria, include those. Otherwise, you can ask the AI to generate suggestions for the evaluation criteria based on the assignment. Tell the AI if there will be a specific number of points, or if the rubric should be based on percentages.

Customize and Refine

Review the AI-generated rubric, adjust criteria or descriptors for clarity and alignment with specific goals, and ensure appropriate language and fairness. Use iterative prompting to have AI assist with the refinement.

Validate and Test

Generative AI can help test the rubric. Prompt the AI to score an example assignment using the rubric. Use the AI feedback to help determine if the rubric is aligned with the objectives of the assignment, if the rubric is clear, and if the scores are reasonable. Calculate the possible points to ensure that satisfactory work will achieve a passing grade.

Finalize and Use

Once the rubric is finalized, AI can continue to assist by analyzing results and suggesting improvements over time.

Prompt Examples

BSN: Clinical Skills Evaluation

Create a rubric for assessing BSN nursing students during a level one clinical skills lab. Criteria include patient communication, adherence to safety protocols, proper technique, and documentation accuracy. The rubric will be worth 100 points. Use a 4-level performance scale (Exceeds Expectations, Meets Expectations, Needs Improvement, Unsatisfactory). Include detailed clinical descriptors for each level. Create the rubric in a table format with the performance scale at the top, criteria on the right-hand side, and the points embedded at each level. Include a word version that can be edited and downloaded.

BSN: Evidence-Based Practice Paper

Generate a rubric for evaluating a BSN-level evidence-based practice (EBP) paper. Criteria include PICOT question formulation, literature review quality, integration of research evidence, and 7th ed APA formatting. Use a 5-point scale and provide specific academic language for each criterion. The paper is worth 50 points.

MSN: Advanced Health Policy Analysis Paper

Develop a rubric for a policy analysis paper for a health policy course written by an MSN nursing student. The assignment will focus on selecting a specific policy, clarity of policy issues, depth of analysis, use of evidence, policy recommendation strength, and scholarly writing using 7th ed. APA formatting. Use a 1–5 scale and align descriptors with graduate-level critical thinking and synthesis skills. The paper will be worth a total of 50 points.

DNP: Systems Leadership Analysis Paper

Generate a rubric for assessing a DNP nursing student systems leadership paper. Include criteria for identifying systems-level problems, application of leadership theories, strategic change planning, and use of quality improvement frameworks. Use a 5-point scale and ensure language reflects doctoral-level

critical thinking and synthesis. Generate and provide specific academic language for each criterion.

Next Steps

Create student-facing rubric explanations or guides to help students understand expectations clearly, improving transparency and supporting self-assessment.

Use AI to adapt a core rubric framework for multiple assignments. For example, modifying criteria for projects that include written papers, presentations, simulations, or clinical performance assessments.

Use AI to create simplified rubrics suitable for students to evaluate their own or their peers' work, fostering reflection and collaborative learning skills.

Develop feedback comment banks linked to rubric criterion and performance level (e.g., "excellent," "needs improvement") that educators can personalize and use to save time when grading.

Use AI to summarize rubric data and explain how rubrics support learning outcomes, which can be included in accreditation self-studies or quality improvement reports.

Map rubrics to course and program outcomes and competencies (e.g., AACN Essentials, QSEN competencies), ensuring strong alignment and evidence for accreditation.

Ask AI to create sample scoring scenarios and practice assignments so educators can align grading standards and improve inter-rater reliability.

Rubric-Aligned Feedback

Generative AI can improve grading efficiency by providing a bank of rubric-aligned comments that educators can use to highlight student strengths and opportunities for improvement. Preparing a comment bank before grading can help educators focus on specific characteristics of high-quality work, helping to avoid reliance on "good work" or other vague feedback. A comment bank for work that needs improvement can help educators provide feedback that is supportive and focused on ways the work can be strengthened.

One drawback of using a feedback bank is that students may feel the feedback is not personalized if educators copy and paste the same feedback into multiple assignments. To prevent this, educators can use the bank as feedback templates, changing the wording for each student to maintain personalization.

Overview of Process

Prepare the rubric

Ensure the rubric provides information about the characteristics of high- and low-quality work for the assignment. Prepare a version of the rubric that can be shared with the AI, such as a pdf.

Specify the comments desired

Provide the rubric to the AI and prompt it to generate student feedback comments based on the rubric. Include information about the number of comments you want, whether you want feedback for high- or low-quality work, and if you want comments to focus on specific criteria.

Edit the list

Review the list and remove any comments that misinterpreted the rubric, are too general to be helpful, or are otherwise unusable. If necessary, prompt the AI to generate additional comments until you have enough comments to complete your grading.

Prompt Examples

Positive feedback

Provide a list of ten comments I can make about student work. The comments should address criteria from the attached rubric and communicate ways the student work is exceptional.

Negative feedback

Use the attached rubric to create five specific comments for students whose assignment needs improvement. Base the comments on the first five criteria, do not include following APA.

Feedback for a Specific Criterion

Attached is the rubric for a student assignment. Use the information for criterion one to write fifteen comments that can be provided to students as feedback about their work. Include at least two comments for each level of the criterion.

Next Steps

Prompt the AI to turn your rubric into a student self-assessment tool. Provide this to students along with the assignment instructions to help students understand how a rubric will be applied for grading.

Keep track of the types of comments you make. Provide AI with the distribution of feedback and the assignment instructions to provide feedback on potential improvements to the instructions or rubric.

Ask the AI to provide feedback based on different patterns of student performance. For example, have AI create comments specifically for students who do well in criteria 1-3, but have problems with 4 and 5. This builds more nuanced responses for common student errors.

Group Feedback

Well-crafted group feedback fosters a sense of collective learning and can prompt students to revisit their own work through the lens of peer contributions. It can also reduce grading fatigue. Generative AI tools can support the creation of group feedback by helping educators articulate patterns, highlight shared misunderstandings, and suggest next steps for improvement. However, to remain FERPA-compliant, educators do not input identifiable student information, direct quotes, or upload completed student assignments into public AI tools. Instead, educators can summarize performance trends in aggregate (e.g., "Several students struggled to connect practice variation to system-level policy"), then use AI to help write polished, instructive feedback. This approach preserves privacy while leveraging AI's strengths in language generation and instructional clarity.

Overview of Process

Note patterns of strengths and weaknesses

While reviewing individual assignments, note patterns of strengths, common misunderstandings, or gaps in learning. Use your notes to create a summary of the topics to include in the group feedback.

Draft and customize feedback

Prompt the AI with your list of strengths and learning gaps to draft feedback. Refine and customize the feedback. This can be completed with the AI or on your own.

Prepare for distribution

Ask the AI to prepare the feedback for the format you will use such as a class announcement, inclusion in a lecture outline, or as a script for a recorded video.

Prompt Examples

Summarizing your feedback notes

Attached are the notes I took while grading student papers. Please analyze my notes and identify the strengths and the gaps in group learning.

General group feedback

Attached are the assignment instructions. Several students successfully applied the Institute for Healthcare Improvement Model for Quality Improvement, but many struggled to write measurable outcome statements. Write group feedback for graduate nursing students that affirms their strengths and guides them to improve outcome specificity. Use a constructive, professional tone.

Feedback on a specific course objective

I just reviewed student submissions for a master's-level assignment where students created change proposals using the Plan-Do-Study-Act model. Most students explained the steps correctly but struggled to connect the 'Act' phase to their plan. Write group feedback that acknowledges strengths and addresses this common gap. Align the feedback with AACN Essential 5.1l 'Evaluates organizational systems that affect care delivery and patient outcomes.' Use a professional, instructive tone and include one reflection question for students.

Feedback with Reteaching plan

I want to provide group feedback and reteach a key concept from a formative assignment on the nursing process completed by pre-licensure BSN students. Most students were able to list components of the nursing process, but many struggled to distinguish between assessment and evaluation. First, generate supportive and encouraging group feedback that acknowledges what students did well and clearly addresses this common misunderstanding. Then, write a brief narrative I can share with students explaining why this distinction is important for understanding nursing interventions and patient outcomes.

Next Steps

After tracking the types of feedback given on recurring assignments, ask the AI to suggest changes to the instructions, rubrics, or preparatory content based on the feedback reports.

As the AI to review your feedback summaries and identify any changes in student performance across learning cohorts. This can help you determine if your instructional changes have had a positive or negative effect.

Use your group feedback to identify ways to improve your course. Ask the AI to summarize common student feedback themes over several weeks of the course to detect learning gaps across the assignments. Ask the AI to make suggestions for your lesson plans based on the feedback themes.

Use your feedback reports to generate a student performance report identifying AACN Essential or program outcomes where students struggled the most. Hand-off the report to the educator teaching the next level so they can increase their focus on those areas.

Turn your feedback summaries into an evidence artifact for accreditation self-studies or annual program reviews by showing how the student data informed course improvements.

Ask AI to generate a reflective teaching summary for your promotion portfolio. To achieve this, ask the AI to summarize how your group feedback demonstrated responsiveness to student learning needs and instructional improvement.

Part Four: Generative AI Applications in Scholarship

Developing Evidence-Based Practice Questions

Using generative AI while developing an evidence-based practice (EBP) question offers nurses a powerful way to structure their thinking and clarify complex clinical problems. Before engaging with AI tools, it is essential to articulate the clinical issue clearly and decide on an appropriate framework, (e.g. PICOT or SPIDER) to guide question construction. Generative AI can support this process by suggesting ways to refine the scope of the question, ensuring it is neither too broad nor too narrow, and prompting nurses to consider alternative angles or overlooked elements. By leveraging AI as a planning partner, nurses can build stronger, more focused questions that address genuine gaps in practice and guide effective evidence searches.

Defining the core components of the EBP question is equally critical. Nurses can use AI to ensure they are maintaining consistent, precise terminology throughout the question. This helps avoid ambiguity and supports rigorous evidence appraisal. Documenting each step and clearly noting how AI suggestions influenced the final question enhances transparency and ensures the process remains reproducible. When used thoughtfully, AI can strengthen the foundation of evidence-based practice and foster deeper clinical inquiry.

Overview of Process

Identify and Clarify the Clinical Problem

Generative AI can support this stage by helping you brainstorm potential problems, rephrase your initial statements to improve clarity, and explore related clinical concerns that may not have been immediately obvious.

Define the Patient Population

AI can assist by generating ideas for potential subgroups, helping you decide on inclusion and exclusion criteria, and suggesting ways to refine or broaden the scope as needed. This collaborative approach helps avoid ambiguity and supports more effective evidence gathering later.

Articulate the Intervention and Comparator

AI can propose alternative interventions or comparators you might not have considered, describe standard practices in various settings, and clarify differences between similar approaches.

Determine the Outcomes of Interest

AI can support this step by listing potential clinical, patient-reported, or system-level outcomes, suggesting which might align best with your goals, and helping prioritize which ones to include.

Refine the Scope and Focus

AI can help analyze different versions of your question for specificity and breadth, provide feedback on feasibility, and recommend adjustments to strike the right balance. This iterative refining process enhances the clarity and utility of your EBP question.

Ensure Language and Terminology Consistency

AI can review your question to identify inconsistencies, suggest standardized vocabulary, and highlight potential ambiguities. This step ensures your question communicates exactly what you intend and aligns with established terminology in the field.

Document the Process Transparently

AI can assist in compiling prompt histories, summarizing iterations, and creating an organized log of changes and rationale.

Example Prompts

Clarifying the Clinical Problem

Help me clarify the main clinical problem related to high rates of hospital readmission in older adults after heart failure treatment. What are some ways to describe this problem clearly and concisely to guide my PICOT question?

Defining the Patient Population

Suggest ways to precisely define the patient population for my PICOT question on using group prenatal care to improve outcomes. What age ranges, risk factors, or settings can I consider including or excluding?

Articulating the Comparator

List possible comparators for evaluating a new telehealth-based diabetes education program in adults with type 2 diabetes. What are standard care

approaches or alternative interventions I can consider using in my PICOT question?

Refining a PICOT Question

Review my PICOT question below and suggest ways to make it more focused. What parts might be too broad or too narrow, and how can I adjust the scope to make it more practical and answerable?

Ensuring Consistent Terminology

Review my draft PICOT question below and suggest any changes needed to use standardized terminology. Identify any ambiguous words or phrases and recommend using MeSH terms or other controlled vocabulary where appropriate.

Next Steps

Use generative AI to explore related sub-questions. Ask the AI to suggest exploratory or supporting sub-questions to develop a suite of related questions that can inform future projects.

Explore your question from different perspectives. For example, use AI to explore the question from a different discipline to identify potential collaborations.

Map your question to funding priorities and strategic goals by asking AI to identify current calls for proposals.

Co-Designed Literature Search Strategy

Generative AI can support scholars in developing literature search strategies. Rather than replacing expert judgment, generative AI acts as a collaborative partner that helps to organize ideas, suggest additional angles, and clarify potential gaps in the initial approach. By leveraging AI's ability to quickly analyze and synthesize large amounts of information, researchers can streamline the process of designing comprehensive and targeted search strategies while still retaining full control over critical decision-making.

Generative AI can help broaden and refine the scope of a literature search. It can generate expansive lists of keywords, synonyms, and subject headings, identify additional relevant databases beyond the most common sources, and suggest alternative concepts to reduce bias and improve inclusivity. AI can also assist in refining search strings based on preliminary results and evolving questions.

Overview of Process

Brainstorm and Expand Keywords and Concepts

Use AI to generate comprehensive lists of keywords and related concepts. AI can help surface synonyms, broader and narrower terms, and controlled vocabulary (such as MeSH terms), reducing the chance of missing important literature.

Identify Appropriate Databases and Sources

Ask AI to recommend relevant databases and information sources. While traditional databases like PubMed or CINAHL are commonly used, AI can suggest additional or specialty databases, as well as grey literature sources, to broaden the scope of your search and include diverse or interdisciplinary perspectives.

Draft Initial Search Strings

AI can provide multiple variations and help you experiment to achieve the right balance between sensitivity (finding all relevant results) and specificity (excluding irrelevant ones).

Test and Refine

Use AI iteratively to analyze results from test searches to recommend adjustments, such as adding new terms or modifying combinations. This cycle of testing and refinement helps ensure that your search strategy effectively captures the evidence you need.

Example Prompts

Generate Keywords

List possible keywords, synonyms, and related terms for my topic on nurse-led interventions to reduce hospital readmissions. Include both general and specific terms.

Suggest Database Specific Vocabulary

Identify potential MeSH terms or CINAHL subject headings for my topic on prenatal education programs for first-time mothers.

Identify relevant databases

Recommend the most relevant databases and grey literature sources to search for studies on telehealth effectiveness in rural primary care settings.

Draft Search Strings

Help me build draft Boolean search strings for this PICOT question [PICOT]. Use my keywords and MeSH terms.

Recommend filters and limits

Suggest appropriate filters or limits (e.g., publication date, language, study type) to apply to my search strategy based on this PICOT question [PICOT].

Next Steps

Use AI to help identify policy documents, reports, theses, and other non-indexed studies (grey literature) that might be missed using a traditional search.

Use AI to help compare search strategies across published systematic reviews and meta syntheses.

Writing Critique

Educators in nursing and other health professions engage in high stakes writing across diverse formats. Each of these requires precision, clarity, and adaptation to different audiences and rhetorical goals. Generative AI offers a responsive, on-demand writing critic that can enhance productivity and improve writing quality. These tools can detect issues in tone, structure, clarity, and coherence to help educator writers identify weaknesses that may go unnoticed in self-editing. By providing suggestions and alternatives grounded in strong writing conventions, large language models serve as a practical extension of educator writing habits, saving time while improving output.

Using AI as a writing critic fosters reflective writing practice and professional growth. Educators can experiment with revising sentences, paragraphs, or entire documents based on AI critiques. Educators can compare versions to strengthen their voice, argumentation, or engagement with readers. This iterative feedback process supports communication, persuasive framing, and alignment with institutional goals. Moreover, the tool can level the playing field for educators who may lack access to writing centers or peer support. As writing expectations in academia shift toward more accessible and impactful communication, AI-powered writing critique offers educator a way to adapt and excel.

Overview of Process

Define Your Purpose

Before using AI, clarify your writing goal. Are you preparing a manuscript, grant, email, syllabus section, or reflective statement? What is the desired tone? Who is the intended audience? Include the purpose in the prompt to improve the AI feedback.

Prepare Your Draft or Excerpt

Format your draft so the AI can read it. This may be by pasting it into the chat or adding it as a readable document.

Prompt the AI to Critique the Writing

Provide the AI with the purpose of the writing, the written draft, and a short description of what you want feedback on (e.g., “Check for clarity and flow”).

You can repeat the process multiple times, asking the AI to focus on a different type of critique each time.

Review and Reflect on Feedback

Read the AI's response critically. Consider that not all suggestions will improve the text. Some changes may distort your voice or intent. Other suggestions may make your writing unclear or overgeneralized.

Explore Comparative Versions

Ask the AI to generate alternative phrasings or restructuring options for language that may be confusing. Version comparisons can help you clarify your preferences and refine your voice.

Revise and Re-test

Apply the changes that resonate with you, then re-submit the revised text for a second round of review. You can ask the AI to complete additional rounds of review by focusing on different aspects of your writing.

Example Prompts

Critique assignment instructions

Act as a writing specialist for nursing education. Review the following assignment instructions for clarity, tone, and alignment with graduate-level expectations. Identify any confusing or overly complex sentences, suggest ways to make expectations more student-centered, and flag any missing components (e.g., due date, submission method, grading criteria). My goal is to ensure students understand what to do and why it matters.

Critiquing a section of a Journal Article Draft

You are a scholarly writing coach. Please critique the following introduction section of a manuscript intended for submission to a peer-reviewed nursing journal. Evaluate the clarity of the problem statement, relevance of the background, logical flow, and effectiveness of the transitions between paragraphs. Suggest ways to strengthen the argument and improve the scholarly tone while preserving my voice.

Improving a Reflective Statement for a Promotion Dossier

I'm preparing a reflective teaching statement for my promotion dossier. Please review the following two paragraphs and give feedback on tone, coherence,

and alignment with common educator evaluation criteria. I want to make sure the narrative sounds authentic and reflective without becoming overly personal or vague. Identify any sections that need stronger evidence of impact or clearer connection to teaching philosophy.

Next Steps

Try multiple generative AI tools (e.g., ChatGPT, Claude, Copilot) to compare writing critiques. This builds your skill in evaluating AI quality and trains you to spot strengths and blind spots in machine-generated advice.

Use AI during writing group writing sessions to prompt real-time revision or compare alternate phrasings. This can enrich peer feedback by focusing on structure, tone, and logic without requiring peers to rewrite each other's work.

Simulated Peer Reviewer

In academic and professional settings, peer review is a cornerstone of quality assurance, yet it often comes late in the development process, after drafts have already circulated or been submitted. Generative AI offers educator a way to simulate the peer review process much earlier, serving as a low-stakes, always-available reviewer. By prompting the AI to take on the role of a journal editor, grant reviewer, or other review committee member, educators can receive structured, targeted feedback that mimics what a real reviewer might say. This early critique can help identify gaps in clarity, logic, alignment with guidelines, or fit with intended audiences.

Using AI as a simulated peer reviewer fosters reflective practice. When used thoughtfully, this approach promotes iterative revision and can save time by reducing the number of revision cycles needed after formal review. It also democratizes access to helpful critique, especially for early-career educators or those without robust mentoring networks. Rather than replacing human review, AI can serve as a rehearsal partner to refine ideas, sharpen their arguments, and prepare their work for successful dissemination.

Overview of Process

Clarify the Purpose and Audience

Before engaging the AI, define what kind of review you're seeking. Is it for a journal article, a grant application, a conference abstract, or a curriculum document? Be specific about the intended audience or organization (e.g., NIH, Nursing Education Perspectives, internal curriculum committee).

Share the Instructions and Rubric

Prepare the submission instructions and scoring rubric for sharing with the AI. This may mean copying and pasting into the chat window or preparing them as a document that the AI can read. Some AI tools may be able to read the instructions from a live webpage.

Request Structured, Category-Based Feedback

Instruct the AI to act as a reviewer and to give feedback based on the instructions and the rubric. This helps avoid vague or generic responses.

Reflect and Revise

Use the AI-generated review as a first draft of feedback. Reflect: Does it highlight anything new? Would a human reviewer say the same? You can prompt AI to revise its feedback to be more detailed or reviewer-style.

Repeat with Fresh Prompts or New Sections

Review other parts of the work or try another "reviewer perspective" (e.g., a skeptical reader, an interdisciplinary reviewer, a novice learner).

Example Prompt

I would like you to act as a simulated peer reviewer for the following [type of document: e.g., journal article, grant application, course syllabus, etc.]. This draft is intended for [target audience or venue, e.g., Nursing Education Perspectives, an NIH R03 grant, internal curriculum committee].

Provide structured, constructive feedback under the following categories:

Clarity of Purpose

Strength and Logic of Argument

Fit for the Intended Audience

Writing Quality and Tone

Suggestions for Improvement

If helpful, you may use a scoring scale (1–5) or note areas that would raise concerns for real reviewers. Please adopt a professional and supportive tone. Here is the text for review:

[Paste your draft section here. Aim for 300–700 words per review round.]

Next Steps

Use AI to simulate reviewers with different tones or levels of expertise. For example, ask AI simulate a novice reader, then act as an expert reader.

Use your AI peer reviewer to compare your draft for alignment against external benchmarks such as reporting criteria, journal submission guidelines, grant scoring rubric, or APA formatting.

Part Five: Generative AI Applications in Simulation

Simulation Scenario Co-Designer

AI can be a powerful partner in co-creating healthcare simulations by streamlining the design process and enhancing realism. The objective of this case is for nurse educators to learn how to collaboratively design a healthcare simulation scenario using AI tools to generate patient cases, develop clinical dialogue, and create supporting materials aligned with defined learning outcomes. While AI enhances the design and delivery of simulations, it works best as a collaborative tool alongside clinical educators to ensure clinical accuracy and meaningful learning experiences.

Overview of Process

Build the simulations piece-by-piece

An iterative, layered approach will be essential in having AI help create a simulation scenario. A simulation scenario is multi-faceted and multi-layered. Trying to add too many inputs into the AI chat will not yield a quality and usable product.

Define your objectives

The first step is to define clear learning objectives, such as improving clinical judgment, communication, or recognizing gaps in knowledge or practice.

Generate potential scenarios

Once the objectives are set, AI can assist in generating realistic patient case scenarios tailored to various conditions, demographics, and acuity levels relevant to the learning gap and capable of eliciting the desired learner behaviors. Nurse educators provide AI with the scenario focus, level of learners, and clinical environment. AI can suggest patient profiles, including background, demographics, chief complaint, and relevant comorbidities.

Determine performance measures

Nurse educators can provide the AI with a list of critical actions the learners should complete during the scenario or use the AI to help create performance measures aligned with specific learning objectives. The AI can suggest key decision points for the performance measures and build these into the scenario.

Align with Standards of Care

Nurse educators can use AI to crosswalk critical actions with existing clinical practice guidelines or patient safety frameworks (e.g., AHRQ TeamSTEPPS, AACN Essentials, Joint Commission standards). This ensures the scenario remains evidence-based and defensible for evaluation.

Script the dialogue

Use AI to script detailed dialogue and branching decision points that mimic real-life interactions with patients, families, or interprofessional teams.

Create any supporting materials

AI can help create supporting materials like pre-briefing guides, performance checklists, and evaluation rubrics aligned with competency-based outcomes.

Build debriefing tools

AI can be used to design standardized tools for debriefing. Alternatively, the AI can assist by analyzing learner responses and providing targeted feedback or identifying patterns that inform future instruction.

Test and Refine

Once the case is designed, AI can be used to simulate learner responses or highlight possible confusion points. Nurse educators can adjust the pacing, embed specific cues, or add complexity to better match learner readiness.

Step by Step Simulation Process

Step 1: Defining Objectives

Identify the Learning Gap

Purpose: Determine what students are struggling with or need to practice more.

- Review clinical evaluations, test results, or feedback from clinical preceptors.
- Talk to faculty or use course outcome data to identify common student errors or weak areas.
- Examples: Inadequate recognition of hypoxia, poor handoff communication, or medication errors.

Develop Clear Learning Objectives

Purpose: Focus the simulation on what the student will be able to do by the end.

- Use SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound).
- Align with Bloom's Taxonomy using action verbs like recognize, demonstrate, prioritize, or intervene.
- Example Objective: *"By the end of the simulation, the student will demonstrate safe medication administration using the six rights."*
- See Writing Module Objectives

Step 2: Design the simulation

Define the Scenario

Purpose: Create a realistic situation that allows the learner to meet the objectives.

- Choose a patient case realistic and relevant to your learning gap (e.g., post-op patient developing hypoxia).
- Include key clinical data: patient history, vital signs, medications, and lab results.
- Example prompt: I'm creating a simulation for novice nursing students. The objective is early recognition of post-op hypoxia. Suggest a realistic case including demographics, surgical history, baseline vitals, and presenting symptoms.

Performance Measures

Purpose: Determine how you will assess whether students meet the objectives.

- List critical actions that align with your objectives.
- Use a checklist or rating scale (e.g., SBAR communication used appropriately, timely recognition of a deteriorating patient).
- Incorporate standards of care and patient safety guidelines (e.g., AHRQ, Joint Commission standards).
- Example prompt: List five critical actions that align with the objective of early hypoxia recognition and safe response. Include a suggestion for how each could be assessed on a performance checklist.

Embedded Challenges and Cues

Purpose: Engage students in critical thinking and test application of knowledge.

- Add purposeful distractions (e.g., a family member asking questions).
- Use confederates (actors or faculty) as techs, providers, or patients to add realism.

- Include decision points where incorrect actions lead to deteriorating patient outcomes unless corrected.

Prepare Equipment and Environment

Purpose: Enhance realism and readiness.

- Work with simulation staff to gather or make supplies and equipment needed for the scenario (e.g., make a 60 ml syringe and create a morphine sulfate label with concentration).
- Create a list for the simulation space that mimics the clinical area as close to reality as possible (e.g., med-surg room, ICU, IV tubing, nasal cannula).
- Create a list of all technology (monitors, IV pumps, call lights) that will be used for the scenario. Ensure everything works properly before running the scenario.

Write Role Scripts

Purpose: Write role scripts using AI to script detailed dialogue and branching decision points that mimic real-life interactions with patients, families, or interprofessional teams. Well-scripted roles make the simulation more realistic and keep it on track. See Co-writing Simulation Scripts.

Patient Script (for standardized patients or voice-over)

- Background: "I had surgery yesterday…"
- Prompts: When to moan, request pain meds, or express confusion.
- Tone/Emotion: Scared, sleepy, in pain, etc.
- Sample: "Nurse, I'm feeling dizzy again. My chest feels tight…"

Family Member / Confederate Script

- Relationship: "I'm her daughter."
- Behaviors: Distracted, panicked, demanding.
- Key lines: "Is she supposed to look like this?" or "I need someone to tell me what's going on!"

Provider Script (faculty or confederate)

- Responds only if contacted.
- Requires appropriate SBAR communication and prompts learners for missing information.
- Uses clinical language: "Order a STAT CBC and troponin."
- Can be used to escalate or de-escalate the scenario based on student action.

Create an Algorithm for Scenario Progression

Purpose: This is a visual guide that can be used by faculty or simulation staff showing how to advance the scenario in response to student actions.

- Use a flowchart or table with three stages:
  - **Initial Condition → Trigger/Student Action → Next Patient Response**
- Include vital sign changes, monitor alarms, or EHR data updates.
- Plan for different branches based on student performance (e.g., if they miss a sepsis cue, the patient deteriorates; if they assess and manage patient with sepsis, the patient reports feeling better).
- See Co-designing the Progression Algorithm.

Sample Progression Flow:

Phase	*Trigger*	*Patient Response*	*Technology Action*
Start	Baseline	Alert but in mild pain	Normal vitals
Mid	Student delays assessment	↑ HR, ↓ BP	Alarm activates, SpO_2 drops
End	Proper intervention	Stabilized	Vital signs normalize

Step 3: Pilot Testing

Purpose: Guide the experience while maintaining a safe learning environment for all faculty and students involved.

- Give a pre-brief that outlines expectations and objectives.
- Let students take the lead; intervene only if necessary for safety or learning goals.
- During the debrief, illicit feedback about the scenario (e.g., what went well, what was confusing and needs clarifying, did the scenario clearly align with the objective?).
- See Pilot Testing the Simulation with Generative AI.

Step 4: Revise based on pilot dry run

Purpose: Revisions based on dry run feedback from faculty and students provide helpful insight about improvements before the scenario is adopted into practice.

- Make updates on the existing patient chart and scenario based on feedback discussed and discovered during the pilot dry run. Be sure to up the same information across all scenario documents (e.g., vital sign changes may appear in algorithm and patient information).

Step 5: Facilitate the Simulation

Purpose: Guide students through the experience while maintaining a safe learning environment.

- Give a pre-brief that outlines expectations, objectives, and the "fiction contract."
- Let students take the lead; intervene only if necessary for safety or learning goals.

Step 6: Conduct a Structured Debrief

Purpose: Solidify learning and promote reflection.

- Use evidence-based models like PEARLS, Gather-Analyze-Summarize (GAS) or additional debriefing frameworks.
- Ask open-ended questions to explore student thought processes.
- Link actions and decisions back to safety protocols and clinical standards.
- See Simulation Debriefing Assistant.

Step 7: Evaluate and Revise

Purpose: Improve the scenario for future use.

- Gather feedback from students and observers.
- Review performance checklists to identify trends.
- Adjust objectives, challenges, or timing based on results.

Co-Writing Simulation Scripts

Generative AI can accelerate and improve the process of simulation script writing by offering structure, inspiration, and iterative feedback. It acts as a collaborative writing partner that supports the educator in designing realistic, pedagogically aligned, and adaptable scenarios. Instead of starting with a blank page, nurse educators can rapidly generate first drafts, test variations, and refine scripts to meet the level, objectives, and context of the learners. AI amplifies clinical expertise.

Overview of Process

Define Scenario Purpose and Learning Objectives

Nurse educators start by clarifying the clinical and educational goals of the simulation. This includes learner level, competency domains, critical actions, and the type of clinical decision-making to be evaluated.

Generate Scenario Framework with AI

Nurse educators provide AI with the core parameters: learner role, setting, key condition or diagnosis, and learning objectives. AI generates a rough script structure: background narrative, patient profile, setting description, and scene flow.

Refine Characters, Dialogue, and Timing

Using AI, nurse educators can co-develop rich character backgrounds, realistic patient or family dialogue, and the sequence of clinical cues (vitals, lab results, changes in patient condition). This includes customizing emotional tones and escalating clinical complexity.

Embed Decision Points and Debrief Anchors

Nurse educators use AI to identify and mark key decision moments that can be unpacked during debriefing. These are tied directly to expected learner actions and competencies. AI can also suggest embedded "failure points" to allow safe exploration of common clinical errors.

Finalize and Format for Simulation Team Use

AI helps reformat the script for standardized patient use, confederate roles, and simulation tech cues (e.g., monitor changes, voice lines). Nurse educators review for fidelity and educational alignment before final deployment.

Example Prompts

Generate a Script Outline

I'm designing a simulation for senior nursing students on sepsis recognition in a post-op patient. Provide a simulation script outline that includes patient background, presenting symptoms, expected learner actions, and embedded cues for deterioration.

Create Realistic Patient Dialogue

Write patient dialogue for a 67-year-old man who is becoming increasingly confused due to undiagnosed sepsis. Include emotional tone shifts, relevant clinical information, and opportunities for learners to ask effective questions.

Add Escalating Clinical Cues

Add a sequence of vital sign changes and nursing charting entries that reflect a patient progressing from early sepsis to septic shock over 15 minutes.

Design a Script for a Communication Scenario

Create a simulation scenario where a novice nurse must call a provider to report a deteriorating patient. Focus on SBAR communication, uncertainty, and provider pushback.

Next Steps

Use AI to generate a series of simulations organized by clinical focus, learner level, or competency domains. Include multiple versions to accommodate variations in learning goals or team roles.

Leverage AI to adapt nursing scripts for interprofessional simulation with medical, pharmacy, or social work students, emphasizing shared decision-making, documentation, and communication touchpoints.

Incorporate social determinants of health and implicit bias cues into characters and scenarios. AI can suggest framing, phrasing, and sociocultural context for these elements.

Experiment with branching logic simulations, where AI helps generate alternate script paths based on learner decisions—ideal for unfolding or virtual scenarios.

Co-Designing the Progression Algorithm

Generative AI can assist nurse educators in building branching logic that reflects clinical reality, supports flexible facilitation, and adapts in real-time to learner decisions. Generative AI can help nurse educators co-create a visual algorithm that outlines the sequence of clinical events and possible patient responses based on learner performance. This algorithm guides simulation staff and educators during the session while preserving educational integrity and fidelity.

Overview of Process

Define the Scenario Flow Structure

AI can assist in rapidly drafting the flow structure. Nurse educators can provide a short description of the scenario, the intended learning objectives, and key decision points. AI can then suggest preliminary flow table triggers.

Generate the Progression Table or Flowchart

Generative AI can use the selected triggers to generate a draft flowchart or decision table outlining how the patient should respond to various learner actions and inactions. Nurse educators can customize the output, refining timing, complexity, and realism.

Incorporate Dynamic Cues and Escalations

AI can suggest realistic changes in vital signs, monitor alarms, and EHR updates based on learner interventions (or lack thereof). For instance, if a student delays sepsis recognition, AI may suggest escalating fever, falling blood pressure, and changes in mental status.

Build in Branches for Learner Variability

Nurse educators can collaborate with AI to create multiple response pathways that account for common learner decisions. For example:

- **Correct action taken** → Patient stabilizes, expresses relief
- **Action delayed** → Mild deterioration, prompting another cue
- **Action missed entirely** → Patient becomes unstable, requiring escalation

AI can help balance realism and teaching value—ensuring that all branches still meet learning objectives without overcomplicating the scenario.

Format for Simulation Staff Use

Once finalized, the algorithm can be exported into a clean visual or tabular format. AI can help translate this into a printable guide or interactive script viewable on tablets or simulation software dashboards.

Example Prompts

Design Escalating Vital Sign Cues

Suggest a series of vital sign changes and alarms over 15 minutes that reflect worsening respiratory failure in a COPD patient if no interventions are taken.

Generate a Scenario Flow Table

Create a simulation progression table for a patient with early sepsis in a med-surg unit. Include three stages of condition changes based on whether learners intervene appropriately.

Suggest Branching Responses

In a scenario involving postpartum hemorrhage, what patient condition changes should occur if the learner delays uterine massage and fails to initiate IV fluids?

Convert a Text Scenario to Flowchart Format

Take this narrative simulation and convert it into a decision-based flowchart with branches based on student performance at key decision points."

Next Steps

Develop AI-assisted templates for common scenarios (e.g., chest pain, mental status changes, falls) that faculty can adapt and use consistently across courses.

Use AI to create annotated versions of the algorithm with facilitator notes, cueing guidance, and timing hints to improve consistency across facilitators.

Simulate learner behavior through AI (e.g., "simulate a student who misses all initial cues") to test how the algorithm responds to poor performance and ensure the case still results in meaningful learning.

Use AI to build maps that link each branch of the scenario to potential debriefing topics, helping facilitators prepare tailored reflection questions for each learner path.

Pilot testing the simulation with Generative AI

Before running a simulation with learners, faculty must ensure the case flows as intended, critical actions are achievable, and cues are delivered at the right time. Traditionally, this requires scheduling a full rehearsal with a mock learner and simulation staff. Generative AI offers a lightweight alternative: simulating the simulation. AI can help nurse educators preview how learners might respond to cues, test the logic of progression algorithms, and flag weak points in case design, all without needing a rehearsal team.

Nurse educators can use generative AI to pilot test their simulation script by role-playing learner interactions, stress-testing decision points, and optimizing scenario flow and timing.

Overview of Process

Simulate Learner Behavior

Nurse educators begin by prompting AI to act as a learner participating in the case. With the scenario description, objectives, and performance expectations as input, AI can simulate responses to patient cues, ask clarifying questions, or even make realistic clinical errors. This helps identify whether learner cues are effective and whether performance measures are observable.

Test Scenario Logic

AI can step through the scenario algorithm, reacting to each condition and cue as a learner might. Nurse educators can ask the AI to take on different roles (e.g., novice nurse, hesitant communicator, fast responder) to see how the scenario holds up across a range of skill levels. This helps reveal logical breaks, unrealistic transitions, or missing data.

Assess Timing and Cue Pacing

By narrating a simulated timeline, AI can help simulate how quickly events should unfold based on learner actions. This allows nurse educators to adjust cue spacing and escalation triggers to better reflect realistic clinical pacing.

Identify Confusion Points

Nurse educators can ask AI to reflect on the experience: What information was unclear? What cues were ambiguous? Where did decision-making feel

unrealistic or forced? This feedback can help refine language, re-sequence events, or clarify handoff information.

Document Suggested Revisions

AI can help generate a revision checklist or highlight flagged sections for further review. Nurse educators can ask AI to format these suggestions into an editable to-do list for updating the scenario before implementation.

Example Prompts

Simulate Learner Performance

Act as a novice nursing student participating in a simulation for early sepsis recognition. Walk through the attached simulation plan, describing what you observe, what actions you take, and how you respond to new clinical data.

Play the Scenario from Multiple Angles

Simulate the same scenario with two different learners: one who responds promptly to abnormal vitals, and one who misses cues. Note how the patient condition changes and what feedback you would expect in the debrief.

Identify Scenario Weak Points

Based on this simulation script, what parts of the case might confuse learners or lead to unexpected decisions? Are there any cues that are too subtle or unrealistic?

Generate a Simulation Debug Checklist

Create a list of revisions to improve this simulation scenario, including pacing of events, clarity of cues, and alignment of patient responses with expected learner actions.

Next Steps

Develop standardized prompts for nurse educators to use when pilot testing new scenarios, including learner personas and pacing guidelines.

Ask AI to generate structured feedback reports based on pilot testing results, including scenario strengths, risks, and suggested revisions.

Simulation Debriefing Assistant

AI offers valuable support to nurse educators during the debriefing phase of healthcare simulations by enhancing the depth and personalization of the process. AI can tailor reflection prompts, encouraging critical thinking and self-assessment. For novice facilitators, AI provides structured guidance through suggested debriefing scripts, cognitive aids, and evidence-based talking points. AI acts as a co-facilitator, amplifying the effectiveness of debriefing by delivering timely insights, supporting learner reflection, and helping nurse educators deliver more impactful, learner-centered experiences.

Nurse educators can use AI-generated insights to facilitate a simulation debriefing that identifies clinical decision-making patterns, highlights areas for improvement, and reinforces key learning objectives.

Overview of Process

Simulation Scenario completion

Key performance data is collected through facilitator observation, digital logs, or an integrated simulation software.

Data input and AI Analysis

Nurse educators input relevant scenario data into the AI platform, including scenario objectives, learner behaviors, timestamps and critical events. AI will analyze the data to identify patterns, deviations from expected actions, and moments of clinical or communication significance.

Generation of Debriefing Aids

The nurse educator can work with AI to generate personalized prompts designed to promote learner self-assessment and critical reflection. AI can also provide guided debriefing scripts based on common debriefing models (e.g. PEARLS) to support less experienced nurse educators.

Debriefing Session Facilitation

AI functions as a co-facilitator, either asynchronously or in real time, offering suggested questions, real time follow ups, thematic insights aligned with learning outcomes.

Post-debrief synthesis and feedback

The nurse educator can use AI to summarize key takeaways from the debriefing, highlighting learner insights, missed opportunities, reinforced competencies

Example Prompts

Generating Debrief Questions

A nurse educator is preparing a debrief session for prelicensure student nurses. The simulation scenario is attached. Provide a list of questions to use during the debrief. Include questions for each of the following categories: Solicit participant reactions, Review the events from the participant's perspective, examine clinical decision making, critique communication with patient and team, and summary thoughts.

Creating a Facilitation Guide

A nurse educator is facilitating a debrief for novice nursing students (R) regarding their simulation scenario they participated in. The debrief period is a time for reflection on student perceptions, behaviors and actions, and evidence of critical thinking (I). Write a facilitator guide to ensure the facilitator can connect application of knowledge and objectives align (S). Questions should be open ended (N). Questions are for novice learners but should vary in complexity from application, evaluation, and synthesis of knowledge (E).

Creating Educator Development Tools

An experienced nurse educator is facilitating educator development for a group of novice educators (R) on how to debrief nursing students. The debrief period is a time for reflection on student perceptions, behaviors and actions, and evidence of critical thinking (I). Write a guide to facilitate the novice educators as they debrief students for the first time in debriefing (S). Include guidelines, summary, and rationale of debriefing best practices (E). These guidelines should be usable for any educators new to simulation and simulation debriefing (N).

Examples of summary or takeaways: What's one thing you learned today that you'll apply in clinical practice? How might this scenario prepare you for real life situations similar to the scenario?

Next Steps

Extend debriefing tools to include interdisciplinary teams (e.g., medicine, pharmacy, social work), using AI to guide discussion on communication, teamwork, and system failures.

Expand the learning by integrate health systems science, policy, and quality improvement elements into debriefings, encouraging students to connect clinical actions with system-level outcomes.

Apply structured debriefing approaches to clinical practicums, community projects, or virtual case discussions to promote reflective practice across the curriculum.

Test virtual debriefing coaches that guide students through reflective questions even outside educator-led sessions.

Use AI to analyze debriefing transcripts or recordings to identify themes, missed learning opportunities, or emotional dynamics, providing meta-feedback to educators and students. Limit this use to AI tools that are FERPA complaint.

Appendix A: AI-Generated Simulation Scenario Template

1. Scenario Title

(Brief, descriptive title)

2. Author(s)

(Your name and any co-authors)

3. Target Learners

(e.g., Pre-licensure nursing students – final semester; New graduate nurses)

4. Estimated Time

- Pre-brief: ___ minutes
- Simulation: ___ minutes
- Debrief: ___ minutes

5. Learning Gap / Rationale

(Describe the knowledge or skill gap this simulation addresses)

6. Learning Objectives (2–4 SMART Objectives)

-
-

7. Scenario Overview

(Brief summary of the scenario setup, patient condition, and major events)

8. Patient Information

- Name:
- Age:
- Gender:
- Diagnosis:
- Past Medical History:
- Allergies:

- Vital Signs at Start:
- Medications:

9. Required Equipment & Supplies

(List all materials and equipment needed, including manikins, monitors, IVs, documentation tools, etc.)

-
-

10. Roles Required

(Include students, faculty, and any confederates needed, e.g., patient, family, nurse, provider)

- Primary Nurse (Student):
- Secondary Nurse/Support (Student):
- Family Member (Confederate):
- Provider (Confederate/Faculty):

11. Environment Setup

(Describe the physical setting—hospital room, clinic, bedside, etc.)

12. Role Scripts

Patient Script

(Used for standardized patient or voice recording)

- Background Story:
- Emotional Tone:
- Verbal Cues to Give:
- Non-verbal Actions:
- Escalation Prompts (e.g., "I feel dizzy" if care delayed):

Family Member / Confederate Script

- Relationship to Patient:
- Key Concerns/Lines:
- Emotional Tone (anxious, angry, calm, etc.):
- Behavior (e.g., asking repeated questions, pacing, calling for help):
- Timing of Entrance or Comments:

Provider Script

- When Contacted:
- Orders to Provide (based on correct communication):
- Optional Escalation (e.g., "Call a rapid" if vital signs worsen):
- Response Variations Based on Student Actions:

13. Scenario Flow Algorithm (Technology Progression)/Events Timeline

(Include planned progression and timing of key events or patient changes. Use this section to guide how the manikin, monitors, and clinical data should change throughout the scenario.)

Phase	*Trigger (Student Action or Inaction)*	*Patient Response*	*Technology Update (Vitals, Alarms, Monitors)*
Start			
Mid			
End			

14. Performance Measures (Checklist or Rubric)

(Include technical and non-technical actions aligned with objectives)

Critical Action	Completed (✓)	Notes
Identifies abnormal vitals		
Calls provider using SBAR		
Administers medication safely		
Documents appropriately		

15. Embedded Challenges & Decision Points

(List challenges or distractions that test judgment, teamwork, or communication)

-
-
-

16. Debriefing Plan

- Debriefing Model Used (e.g., PEARLS, GAS):
- Key Questions to Ask:
    -
    -
- Points to Emphasize (e.g., safety, communication, priorities):
- Link back to objectives:

17. Evaluation & Notes for Future Use

(Instructor notes on what worked, what to change, feedback highlights, etc.)

-

Appendix B: Simulation Scenario AI-Generated Checklist

Step 1: Identify the Learning Gap

- ☐ Review course evaluations, clinical feedback, or student performance data
- ☐ Identify a common knowledge or practice gap
- ☐ Define the clinical focus (e.g., communication, safety, assessment)

Step 2: Develop Learning Objectives

- ☐ Write 2–4 SMART objectives
- ☐ Use action verbs aligned with Bloom's Taxonomy
- ☐ Ensure each objective links directly to an identified gap
- ☐ Example: "Student will conduct a focused assessment for a deteriorating post-op patient."

Step 3: Design the Scenario

- ☐ Create a patient case that matches the learning objectives
- ☐ Include realistic patient data: age, diagnosis, history, medications, labs
- ☐ Design scenario flow: introduction → clinical event → resolution
- ☐ Decide if using a manikin or standardized patient

Step 4: Create Performance Measures

- ☐ List key actions students should perform
- ☐ Develop a checklist or rubric with measurable behaviors
- ☐ Align actions with national safety/clinical standards (e.g., SBAR, Six Rights of Med Admin)
- ☐ Include both technical and non-technical skills

Step 5: Embed Challenges and Cues

- ☐ Add realistic obstacles (e.g., family interruptions, alarms)
- ☐ Plan clinical decision points (e.g., worsening vitals)
- ☐ Use confederates (actors/faculty) to portray staff or family
- ☐ Ensure cues guide learners without giving answers

Step 6: Prepare Equipment and Environment

- ☐ Set up space to replicate the clinical setting
- ☐ Gather all necessary supplies (PPE, meds, IV pumps, etc.)
- ☐ Test manikins, monitors, and other tech for functionality
- ☐ Prepare documentation tools (charts, MAR, EHR mockups)

Step 7: Write Role Scripts

Patient Script

- ☐ Include background info, key lines, and emotional tone

- ☐ Define how the patient will respond verbally and nonverbally
- ☐ List timing for symptoms, pain, or confusion cues

Family Member / Confederate Script

- ☐ Define relationship and personality (e.g., concerned spouse, anxious parent)
- ☐ Include scripted lines and cue timing
- ☐ Add realistic interruptions or questions

Provider Script

- ☐ Include appropriate responses to SBAR communication
- ☐ Define when to offer orders, clarification, or pushback
- ☐ Use to escalate or redirect scenario flow

Step 8: Create a Scenario Flow Algorithm

- ☐ Build a flowchart or table to guide scenario stages
- ☐ Include: patient status → student action → result
- ☐ Plan for branching paths depending on student decisions
- ☐ Indicate when to activate monitor/alarm changes or update vitals

Step 9: Pilot the Simulation Scenario

- ☐ Provide a clear pre-brief (objectives, roles, confidentiality)
- ☐ Set the expectation for psychological safety and realism
- ☐ Let students lead; intervene only for safety or major learning points
- ☐ Collect feedback from participants and observers
- ☐ Review performance outcomes for consistency with objectives
- ☐ Note areas for improvement (timing, cues, realism)
- ☐ Revise the scenario and documentation as needed

Step 10: Facilitate the Simulation

- ☐ Provide a clear pre-brief (objectives, roles, confidentiality)
- ☐ Set the expectation for psychological safety and realism
- ☐ Let students lead; intervene only for safety or major learning points

Step 11: Debrief the Scenario

- ☐ Use a structured model (e.g., PEARLS, GAS)
- ☐ Ask open-ended questions to prompt reflection
- ☐ Highlight actions that promoted or compromised safety
- ☐ Reinforce connection to learning objectives and clinical practice

Step 12: Evaluate & Revise

- ☐ Collect feedback from participants and observers
- ☐ Review performance outcomes for consistency with objectives
- ☐ Note areas for improvement (timing, cues, realism)
- ☐ Revise the scenario and documentation as needed

Part Six: Facilitating Student Use of Generative AI

Beyond AI-Detection: Engaging students with AI

Many nurse educators have responded to generative AI's emergence with caution, focusing heavily on detecting AI-generated work rather than exploring its potential to deepen learning and prepare future nurses for complex clinical realities. This chapter challenges the prevailing "policing" approach and invites nurse educators to adopt an "engagement" mindset. The goal is to see AI as a teaching partner that can expand critical thinking, creativity, and professional readiness.

Relying on AI detection tools may offer a temporary sense of control but ultimately misses the larger educational opportunity. Engaging students with AI equips them to navigate the ethical, practical, and clinical implications of these technologies they will inevitably encounter in practice. Reframing our approach to AI from detection to engagement aligns with the core values of nursing education: fostering inquiry, encouraging reflection, and building professional competencies that extend beyond technical tasks. In this chapter, we will explore how educators can prepare themselves to guide this change, and practical strategies to integrate AI in ways that support student growth and uphold the integrity of the nursing profession.

Rationale for Change

Artificial intelligence is no longer a futuristic concept confined to research labs; it is an active participant in daily clinical practice. AI-driven clinical decision support tools guide medication dosing, identify early signs of sepsis, and flag deteriorating patients. Predictive analytics help allocate resources and reduce hospital readmissions. In outpatient settings, AI tools support remote monitoring, personalized education, and triaging patient messages. For today's nursing students, these tools will be integral to delivering safe, efficient, and patient-centered care. Embracing AI in education mirrors what students will encounter in practice and positions them as confident, capable clinicians rather than passive recipients of technology.

The American Association of Colleges of Nursing (AACN) Essentials emphasizes informatics and healthcare technologies as fundamental to advanced nursing practice AI literacy can be defined as the ability to understand, evaluate, and appropriately use AI tools. Beyond technical skills, AI literacy includes critical appraisal of AI outputs, awareness of algorithmic bias, and knowledge of safe integration into clinical workflows. Without foundational knowledge and hands-

on experience, students may become either overly reliant on or unjustifiably skeptical of AI-driven tools. By fostering AI literacy, educators prepare students to engage with new technologies thoughtfully, advocate for patient safety, and contribute to shaping future technology use in clinical settings.

In educational contexts, AI is already transforming how student nurses learn. Adaptive learning platforms personalize content delivery, virtual simulations replicate high-stakes scenarios, and AI chatbots guide students through complex case studies. A compliance-focused mindset that emphasizes detection of and punishment for generative AI use inadvertently promotes surface learning. When students fear being "caught," they may avoid risk-taking and fail to develop deep understanding. In contrast, when AI is used as a learning partner, it offers opportunities for students to apply, analyze, evaluate, and even create knowledge.

For example, students can use AI to draft patient education materials and then critique the output for accuracy and cultural sensitivity or simulate policy briefs and identify weaknesses. These activities foster critical engagement, not blind acceptance. By shifting from a focus on detection to a focus on higher-order thinking, nurse educators can nurture curiosity and clinical reasoning skills.

Integrating generative AI into learning activities

Once an educator adopts a partnership mindset and builds foundational skills, the next step is to design practical, engaging learning activities that incorporate generative AI in meaningful ways. These activities move beyond passive exposure to technology and actively develop students' critical thinking, clinical reasoning, and professional judgment.

For example, in a health policy course, students can use an AI tool to draft an initial policy brief on a nursing workforce issue. Then, students critically analyze and revise it, comparing AI suggestions with evidence from current literature and regulatory standards. In a pathophysiology course, students could use AI to generate differential diagnoses for a simulated patient scenario, then critique the accuracy and discuss potential risks of over-reliance on AI in clinical decision-making.

Importantly, these activities will emphasize transparency: students can explicitly state when and how they use generative AI, fostering habits of disclosure and integrity. Educators can create assignment rubrics that include criteria for

reflecting on AI use, evaluating its limitations, and demonstrating critical appraisal skills.

As nurse educators integrate AI into coursework, they remain vigilant about equity and access. Not all students have equal familiarity with or access to advanced AI tools. Differences in technology literacy, financial resources, and even language fluency can shape how comfortably students engage with AI-supported assignments.

Strategies to promote equitable access include:

- Providing institutional access to AI tools so students are not required to pay for subscriptions or rely on free but potentially lower-quality services.
- Offering structured orientation sessions to introduce AI basics and discuss responsible use, creating a shared foundation.
- Building peer mentoring opportunities, where more tech-confident students can support those with less experience.
- Designing inclusive assignments, ensuring that AI integration enhances rather than hinders participation for all students.

Educators are attentive to the risk of reinforcing bias. AI systems can perpetuate and amplify existing inequities if not critically evaluated. Assignments can encourage students to question outputs for cultural sensitivity, representation, and ethical implications. This practice builds both AI literacy and social accountability which are crucial for nurses serving diverse populations.

Fostering Professional Accountability

At its core, nursing education is about preparing practitioners who act with accountability and integrity. Introducing AI as a co-learning tool is an opportunity to teach students that using new technologies responsibly is part of their professional identity.

This involves establishing clear expectations: students will explicitly disclose when and how they use AI tools, whether drafting patient education materials or assisting in reflective writing. Educators can reinforce this by requiring written AI use statements in assignments and modeling openness about their own use of technology in teaching.

Such transparency mirrors what will be expected in clinical practice. Just as nurses disclose when they use decision-support algorithms or follow specific protocols, future nurses will need to communicate openly with colleagues and patients about how AI informs their decisions.

Nursing students are accustomed to working within ethical frameworks such as the ANA Code of Ethics. By inviting students to reflect on and develop personal AI use guidelines, educators empower them to carry ethical reasoning into uncharted technological territory.

Class activities might include:

- Creating a personal AI use code, articulating individual principles for when and how they will engage with AI.
- Group discussions on case studies where AI-generated decisions led to ethical dilemmas in healthcare.
- Comparing institutional honor codes with emerging best practices for academic and clinical AI use.

Through these exercises, students learn to balance innovation with caution, understand potential harms of algorithmic bias, and develop the moral courage to question or refuse technology that conflicts with patient-centered values.

Addressing Algorithmic Bias and Equity

Ethics in AI is not limited to disclosure and personal responsibility. It also demands vigilance in identifying and addressing bias within AI systems themselves. Many AI tools are trained on datasets that may lack diversity or perpetuate existing health inequities. For example, risk prediction algorithms that underrepresent certain populations can lead to unequal treatment recommendations.

Forward-thinking healthcare organizations increasingly expect nurse leaders to guide the safe and strategic adoption of emerging technologies. This requires a blend of clinical insight, operational understanding, and technological literacy. Future nurse leaders will need to:

- Evaluate AI tools for clinical appropriateness and alignment with patient-centered care.
- Identify opportunities for AI to improve efficiency, reduce burnout, and enhance patient outcomes.
- Advocate for equitable and ethical implementation of AI across diverse patient populations and communities.

By embedding these skills into nursing education now, educators empower students to step confidently into leadership roles that shape technology adoption rather than merely react to it. Educators can support this by including assignments that require students to propose improvements to current workflows using AI. Educators can encourage students to critique existing AI-driven solutions and suggest alternatives. When feasible, offer cross-disciplinary collaborations with engineering, computer science, or data science students to co-create innovative health solutions.

Ultimately, leadership in AI is grounded in nursing's enduring mission: to advocate for patients and communities. As technology advances, patients will increasingly rely on nurses to interpret, question, and humanize AI-driven recommendations. Nurse leaders will play a critical role in ensuring that technological innovations do not compromise dignity, equity, or trust in care relationships. By engaging students in these discussions, educators help cultivate ethically grounded, patient-focused leaders ready to influence the future of healthcare policy, technology development, and clinical standards. Such experiences reinforce the idea that nurses are essential system actors who shape care delivery at every level.

Maintaining Critical Thinking When Using AI

Cognitive load and automation bias

Nurses work in environments brimming with complexity. Each day they juggle lab results, patient histories, medication regimens, documentation demands, and care coordination. This mental juggling act reflects what cognitive scientists call cognitive load. When cognitive load becomes too high, our capacity for critical thinking is compromised in subtle but impactful ways.

Cognitive load can be divided into three types. Intrinsic load refers to the inherent difficulty of a task itself. Extraneous load arises from distractions or poorly designed systems that clutter thinking. Germane load, on the other hand, supports learning and problem-solving, the mental effort that deepens understanding and builds expertise. Ideally, AI can help reduce extraneous load and enhance germane load, freeing nurses to focus on patient care and decision-making. Yet, when misused, AI can introduce new risks including automation bias.

Automation bias is the human tendency to place excessive trust in machine-generated recommendations, particularly when under stress, fatigue, or time pressure. Imagine preparing for a critical care planning meeting. You ask an AI to summarize an important article, and it returns a polished summary that feels complete. Yet it omits a crucial recommendation — and you miss it, not out of carelessness, but because the AI's authoritative tone convinces your brain that the thinking is already finished. This is automation bias at work: the mind's instinct to conserve effort leads us to accept machine outputs uncritically.

Strategies to Support Critical Thinking

To counteract automation bias, nurses can adopt simple but powerful strategies, such as requesting information in bullet points or numbered lists. This formatting forces active engagement rather than passive acceptance, prompting us to interpret, question, and integrate new information consciously. By insisting on thinking with the AI rather than letting it think for us, we reclaim agency and sharpen our clinical reasoning.

Effective prompting

A crucial skill in working effectively with AI is prompting. The way we communicate with AI systems profoundly influences the quality of their

responses and our own thought processes. Poorly crafted, vague prompts tend to yield superficial or inaccurate answers. Well-structured prompts, however, do much more: they can guide, clarify, and even challenge our own reasoning.

Effective prompting is not merely about efficiency; it is about ownership of thought. For example, rather than telling an AI to "summarize this article," you might ask it to "list three key takeaways and explain how each relates to maternal health policy." You could also ask for multiple perspectives, such as "provide two ways to explain this topic to a patient with low health literacy." Including prompts to specifying sources, timelines, or prohibiting speculation ensures the output aligns with your standards. In this way, the act of prompting becomes a reflective practice, cultivating critical appraisal skills and intellectual discipline.

Matching the task to the tool

Choosing when to use AI is just as important as knowing how to prompt it. Not every task benefits equally from AI support. A helpful framework proposed by Ethan and Lilach Mollick in Co-Intelligence suggests using a "traffic light" system. Green light tasks include brainstorming, outlining, or summarizing one's own notes. These low-stakes, high-utility activities are where AI shines. Yellow light tasks, like summarizing new research or drafting patient materials, require cautious human oversight and thorough verification. Finally, red light tasks such as clinical recommendations, policy evaluations, or legal decisions demand human leadership and remain firmly in our hands. Using AI to summarize a Medicaid policy brief might seem harmless, but if it misses a critical coverage change, the consequences could be serious. In these contexts, AI is a collaborator, never the final authority.

Critical Appraisal of AI Outputs

Even when using AI for lower-stakes tasks, educators engage in critical appraisal. Generative AI is not designed to discern truth; it predicts what sounds plausible based on patterns in its training data. This means outputs can look credible yet contain fabricated information, so-called "hallucinations." Applying tools like the CRAAP Test can help detect these pitfalls. The CRAAP test was developed in 2004 by Sarah Blakeslee and librarians at CSU to help people evaluate online content and can be applied to AI outputs. CRAAP stands for Currency, Relevance, Authority, Accuracy, and Purpose. For example, a beautifully formatted reference list generated by AI may contain entirely

fictitious citations. Beyond technical accuracy, be vigilant about inclusivity and bias. Does the output reinforce stereotypes? Does it marginalize certain populations? Before accepting AI-generated content, every nurse can ask: Would I confidently sign my name to this? If the answer is no, the output is not ready.

Debriefing and Reflection

After using AI, reflective practice becomes essential. Reflection transforms an interaction with AI from a transactional moment into a learning opportunity. Consider a postpartum unit that implemented AI-drafted discharge summaries. One day, a pharmacist noticed that a crucial medication instruction was missing. Rather than blaming the tool, the team re-examined their prompts and processes, adding review steps and enhancing their checks. By pausing to ask what went right, what went wrong, and what to change, they strengthened their system and deepened their understanding.

Human-AI Teaming

At the heart of AI integration is the principle of human-AI teaming. Human-AI teaming is a partnership where technology supports but never replaces human judgment. For nurse leaders, this means modeling sound decision-making, setting clear expectations, and fostering a culture of critical engagement. Just as we would not accept vital sign reading without context, we do not accept AI outputs without scrutiny. Nurse leaders champion thoughtful use, guide teams in evaluating outputs, and speak up when something feels amiss. Leadership in this era involves curiosity, accountability, and unwavering commitment to patient-centered care.

Ultimately, successful use of AI in nursing is not about mastering every tool or chasing every new feature. It is about cultivating the judgment to know when to engage, how to question, and where to draw the line. Our most valuable assets remain our professional expertise, ethical compass, and dedication to safe, compassionate care.

AI can help us achieve more if we lead the partnership with integrity and discernment. In this evolving landscape, it is not the AI that ensures care remains safe and person-centered. It is us.

Key Decisions when Designing AI Interactions

Pedagogical Alignment

AI interaction will work better if educators can give the AI and the students a narrow focus. Consider how alternative perspectives help meet the goals of the course. It is also important to determine what types of perspectives you want students to explore. These might include ethical dilemmas, culturally diverse patient experiences, differing policy stances, or the priorities of various system actors such as administrators, policymakers, or patients. Being clear about the intended focus helps ensure the assignment is meaningful and integrated rather than an add-on activity.

Student Readiness

Educators assess whether students are prepared, both cognitively and emotionally, to engage in this type of assignment. Students need to have a foundation of content knowledge to interpret and analyze perspectives critically rather than simply accepting AI-generated ideas at face value. In some cases, it might be most helpful to interact with the AI as part of a classroom discussion. Before assigning AI interactions as homework, educators can consider whether students have developed enough reflective capacity and psychological safety to process potentially uncomfortable or conflicting viewpoints constructively.

Prompt Design

When designing an AI interaction, creating a clear and specific prompt is one of the most critical steps. The quality of the student experience depends heavily on the detail and direction included in the prompt. Educators will decide how focused the scenario need to be to elicit realistic, meaningful responses from the AI. For example, a vague prompt might lead to generic or unhelpful dialogue, while a well-crafted prompt can simulate the nuance of a patient refusing treatment, a family navigating an end-of-life decision, or a policymaker explaining funding cuts. Educators will also to choose which perspectives the AI will take on or provide. Including potential conflicts, emotional dynamics, and ethical dilemmas in the prompt ensures that students practice navigating real-world complexity rather than simply rehearsing rote communication.

Facilitation Structure

A successful student AI interaction requires careful planning around how much structure and guidance students will receive. Educators decide whether to provide highly guided frameworks such as step-by-step conversation scripts or specific questions to ask or allow more open-ended exploration to encourage critical thinking and creativity. Additionally, educators plan how they will facilitate the learning process: Will the activity be done live in class with immediate coaching? Will students complete it asynchronously with follow-up debriefs? Will small groups share experiences and discuss challenges together? By thoughtfully designing the guidance and facilitation structure, educators ensure that students remain supported throughout the exercise, engage deeply with the scenario, and process the emotional and ethical dimensions in a safe, reflective environment.

Resource and Feasibility Considerations

Educators evaluate whether they have access to a reliable and appropriate AI tool capable of generating high-quality, nuanced perspectives. It is critical to ensure the tool complies with institutional policies and meets privacy and security standards (for example, FERPA or HIPAA considerations where applicable). Time and educator capacity are also important. Consider whether you can dedicate sufficient class or discussion time to prepare students, facilitate engagement, and debrief after the activity. Educators plan for providing guidance materials such as reflection prompts or analysis frameworks to help students navigate the assignment thoughtfully.

Support and Facilitation Planning

Providing clear instructions and structured prompts will help students engage critically rather than passively consuming information. Educators prepare to facilitate discussions, including addressing emotional responses and managing any conflicts that arise. Planning for how to create a safe and respectful learning environment is key to ensuring the activity supports rather than undermines students' growth. Educators decide whether the assignment will be formally graded or used as a formative learning opportunity. Clear criteria for assessing student engagement, depth of analysis, and reflection will help reinforce learning objectives and provide meaningful feedback.

AI-integration and Student Assessment

Educators have a responsibility to lead assessment practices that are both pedagogically sound and ethically transparent, especially when integrating generative AI into assignments. This includes determining how the AI-integrated activity fits within the course's overall assessment strategy. Assignment instructions can communicate whether the task is formative (a practice opportunity) or summative (graded), and clarify how student performance will be evaluated. Rubrics can specify which aspects must demonstrate the student's own reasoning, such as depth of analysis, quality of reflection, communication skill, or ability to consider alternative perspectives.

Students may assume AI use is allowed across all assignments unless told otherwise. To prevent confusion, educators can state clearly for each assignment whether AI use is permitted, expected, or restricted. Communicating these expectations in advance helps students focus on learning objectives rather than guessing the rules. One way to model transparency about AI use is through communicating to students when generative AI tools were used to create learning materials. Nurse educators can also communicate how feedback will be provided, reinforcing the educational value of the activity and supporting student development.

AI-Generated Alternative Perspectives

Integrating AI into student learning as a tool for exploring alternative perspectives offers transformative potential in health professions education. At its core, this approach empowers students to engage deeply with values clarification, encouraging them to critically examine their own assumptions, identify biases, and refine their professional identity. When used thoughtfully, AI-generated perspectives can act as a "thinking partner," simulating viewpoints that students might not otherwise encounter in their academic or clinical environments.

A central goal in nursing education is to prepare students to provide equitable, person-centered care. By interacting with AI-generated scenarios or counterarguments, students can confront unfamiliar cultural, ethical, or policy positions in a structured and safe environment. This process helps them learn to recognize how personal and systemic biases can shape clinical judgments and policy preferences. As a result, students become more aware of the need to adapt their practice to be more inclusive, responsive, and attuned to diverse patient needs and social contexts.

This strategy also helps students understand the complex interplay of priorities and incentives that drive differing positions on clinical treatments, public health interventions, or policy decisions. AI can simulate the reasoning of various system actors, from patients and families to policymakers and administrators. A good AI interaction can make visible the tensions between cost, access, cultural values, evidence, and safety. Through this exploration, students develop systems thinking skills, learning to appreciate why reasonable people can arrive at different conclusions based on their roles and incentives.

To realize this value, educators thoughtfully align AI-based perspective-taking activities with course objectives, consider student readiness, and facilitate reflective processing. When implemented effectively, using AI in this way strengthens students' capacity for self-awareness, empathy, and critical analysis. It helps build a workforce better prepared to advance health equity, navigate policy complexity, and foster inclusive environments.

Potential Applications

Case Expansion

Educators can provide students with a clinical or policy case scenario (for example, the use of life-support in terminal illness). Students will ask AI to generate multiple perspectives reflecting different ethical principles such as patient autonomy, distributive justice, beneficence, and non-maleficence. Students analyze each perspective, identify ethical tensions, and write a reflection or position statement. This helps students appreciate the complexity of ethical decision-making and understand the values that inform different stakeholders' positions.

Decision and Consequences

Educators provide students with a scenario where a decision needs to made. The students will decide and ask AI to suggest the consequences of the decision. For example, choosing to honor a patient's refusal of treatment might lead to family distress, legal challenges, or unexpected clinical outcomes. If students are proficient in AI, they can be asked to progress the scenario through multiple decision points. Students will justify their decisions at each step, reflecting on ethical frameworks and potential trade-offs. This approach highlights the ripple effects of ethical decisions in real-world healthcare contexts.

Debate Partner

Educators can design an exercise where students interact with an AI that takes on a specific ethical stance (e.g., prioritizing individual autonomy vs. public health protection in vaccine mandates). The student will argue their own position and respond to AI-generated counterarguments. This approach challenges students to articulate their reasoning clearly, anticipate opposing viewpoints, and consider the strengths and weaknesses of different ethical frameworks (such as deontology vs. utilitarianism).

AI-Generated Patient Stories

Educators can assign students to create an AI-generated "patient narratives" describing why a patient chose or refused certain treatments. Students then write reflections exploring how they would respond, what biases they might hold, and what communication strategies they would use.

Designing the Assignment

Choose or create your scenario

Select a scenario or decision point relevant to your content area (e.g., a patient refusing treatment, a controversial policy stance, an ethical argument). Depending on your application, this may be a detailed scenario that students include in a prompt or a set of criteria that the AI will use to create a scenario.

Decide on the Student and AI Roles

Define the role of the AI and the role of the student. Provide clear instructions about what students and the AI can or should do (e.g., "Explain how you would respond to this patient," "Argue for or against this perspective," "Reflect on how this changes your understanding").

Write the AI Prompt

Write the prompt by drafting, testing, and refining until you achieve a prompt that provides the expected result. Be sure to test the prompt with inappropriate answers. If students may use different AI tools, test the prompt on different platforms.

Write your Instructions and Include the Prompt

Follow your standard format for giving instructions, including the AI prompt that students will use. Ask students to copy and paste the prompt into the AI chat box to begin the assignment. If you have access to an AI that allows you to build a custom GPT or bot, you can provide a link to the custom tool.

Grading Structure

Determine how to incorporate the AI work into your course grading structure. Educators can ask students to copy and paste the chat experience into a document to submit. Alternatively, students can write a reflection about the AI experience to discuss how they applied course concepts to the AI discussion.

Example Assignments

Students write a prompt to explore perspectives on policy change

Imagine you are preparing for a community forum where diverse stakeholders will discuss a new state health policy proposal (for example, a policy affecting

healthcare access, public health measures, or professional practice regulations).

Ask the AI to provide you with six simulated participants; three that support the proposal and three that oppose it. Ask the AI to give you backstories for the participants and provide information about actions they are taking to promote or block the proposal.

After reviewing the AI-generated perspectives, prepare a three-page discussion of the different perspectives. Consider:

- The values or priorities that might drive support or resistance.
- The potential benefits and harms seen by different groups (e.g., patients, professionals, policymakers, community organizations).
- How these perspectives might reflect broader social, economic, or ethical considerations.
- How this information can be used to design a population health education campaign.

Educators provide a prompt so students can explore patient perspectives

Step 1: Copy and paste the following prompt into your AI tool or chat assistant:

Prompt to copy and paste:

"Give me three different patient perspectives on treatment options for cancer. Provide one patient who wants aggressive treatment regardless of side effects, one who prefers a focus on quality of life even if it shortens survival, and one who is undecided and fearful of making the wrong choice. For each, describe their main concerns, values, and questions they might have for their healthcare team."

Step 2: Review and reflect

- Carefully read the AI-generated responses for each patient perspective.
- Think about what values and emotions are driving each patient's choices.
- Consider how these perspectives would affect the way you communicate, provide information, and build trust.

Step 3: Write your application paper

Write a short application paper (approximately 2–3 pages) that addresses the following for each patient perspective:

- Summary of the perspective: Briefly describe the patient's values, concerns, and questions in your own words.
- Approach to patient education: Explain how you would tailor your education and communication strategies to support this patient's decision-making process. Include how you would address fears, clarify risks and benefits, and promote shared decision-making.
- Reflection: Discuss what you learned about your own communication style and potential biases through this exercise.

Tips for use

Encourage active questioning, not passive spoon-feeding

Students can be guided to ask follow-up questions, challenge explanations, or request clarifications. Remind them: "Ask 'why' like an annoying toddler." Passive consumption won't build critical thinking.

Monitor and reflect

Assign reflective exercises: "Summarize what you learned from the AI tutor. What did you fact-check? What surprised you?" This builds metacognitive skills and prevents mindless trust.

Design check-in points with educator

Don't let the AI tutor replace human guidance entirely. Schedule regular touchpoints so students can discuss what they've learned and clarify misunderstandings before they snowball into exam-day disasters.

Generative AI as a Role-Play Partner

The use of generative AI as a role play partner in health professions education offers a flexible way to strengthen students' communication skills, clinical reasoning, and confidence in patient-centered care. By engaging in simulated conversations, students can safely explore complex scenarios and practice responding to diverse perspectives without the risk of harming real patients or colleagues. When thoughtfully implemented, this approach aligns directly with key educational objectives, preparing students to navigate difficult conversations and support diverse patient decisions in clinical practice.

Creating effective AI prompts is central to realizing these benefits. Prompts are carefully designed to reflect realistic, nuanced scenarios that align with course learning outcomes. For example, prompts might simulate a patient refusing life-saving treatment, a family advocating for aggressive care, or a policymaker explaining a controversial health policy. Clear, structured instructions guide the AI to generate meaningful responses and ensure that students engage actively rather than passively. Well-crafted prompts help bridge the gap between theory and practice, making learning more engaging and authentic.

Equally important is preparing students for this type of learning activity. Students will develop digital literacy skills to interact with AI thoughtfully and critically rather than accepting its outputs at face value. Educators provide guidance on how to approach role play, monitor the quality of interactions, and lead reflective debriefings to process insights and emotional responses. This preparation ensures that students are not only technically ready but also emotionally and intellectually equipped to benefit fully from the exercise.

Despite its potential, using AI as a role play partner also carries important risks. AI-generated responses may be biased, oversimplified, or emotionally challenging for students. Educators work closely with institutional leaders to ensure that appropriate data privacy safeguards and policies are in place before implementation. Institutional support is vital for technical reliability, compliance with privacy standards, and building a culture of ethical innovation. Careful planning around these risks protects students and maintains the integrity of the learning experience.

Designing the Assignment

Define the Scenario Focus

Decide on a realistic context or situation that will allow your students to practice or demonstrate specific competencies. Determine the level of complexity appropriate for the student stage and create objectives for the scenario that align with your course and overall program goals.

Define the student and AI roles

Determine the role the student and AI will play in the scenario. This may be specific roles you assign, or you may give the AI a group of roles to select from randomly. Consider whether the student will lead the discussion and prompt the AI to respond, or if the AI will lead the discussion and ask the student to respond.

Design the AI Prompt

Use your context and roles to write a clear, detailed prompt that guides the AI to generate nuanced, context-specific responses. Be sure to specify how the AI and student roles will be assigned, and how the AI should prioritize responses. Test the prompt and make refinements as necessary until it works as expected.

Plan for Student Preparation

Identify the background knowledge, pre-readings, or frameworks the students need before engaging with the AI role-play. Provide instructions on how to interact with the AI thoughtfully and critically, including guidance on evaluating the AI-generated perspective.

Define your Assessment and Feedback Approach

Decide if the activity will be graded or ungraded. Develop clear criteria for evaluating student engagement, analysis, communication style, or reflection. Plan how and when feedback will be delivered.

Example Assignments

Communicating a Medication Error

Scenario

You are a nursing student completing a clinical rotation on a medical-surgical unit. While preparing to administer medications, you notice that a new prescription for a patient includes a dose that is much higher than what you

learned is safe for this medication. You are unsure if this is intentional or an error, but you feel concerned. You need to clarify this with the physician before giving the medication.

Instructions

Step 1: Prepare

Review your communication tools (e.g., SBAR: Situation, Background, Assessment, Recommendation; or CUS: I'm Concerned, I'm Uncomfortable, this is a Safety issue).

Reflect on why it might feel difficult to speak up and why it is important for patient safety.

Step 2: Use AI to simulate the conversation

Copy and paste this prompt into your AI tool or chat assistant:

AI Prompt to copy and paste:

"Act as a physician on a busy medical-surgical unit. A nursing student approaches you with a concern about a medication dose you ordered, which they believe might be too high. You are initially in a hurry and slightly dismissive because you have many patients to see. Respond as this physician, expressing your initial reactions and responses."

Engage in a short (5–7 minute) written or spoken interaction. Focus on expressing your concern clearly and respectfully, using the communication tools you reviewed.

Step 3: Write a reflection summary (1–2 pages)

In your summary, include:

- Brief description of the interaction: What did the physician say, and how did you respond?
- Your feelings: How did it feel to speak up? What made it difficult or easier?
- What you learned: What strategies helped you communicate effectively? What would you do the same or differently in real life?
- Your role as a student: Describe how this exercise helped you understand your responsibility in patient safety.

Addressing Patient Request for Unnecessary Antibiotics

Scenario

You are seeing a 42-year-old patient presenting with nasal congestion, sore throat, and a mild cough for four days. The patient insists on receiving an antibiotic, believing it will help them "get better faster" because that worked in the past. Based on your assessment, you determine this is a viral upper respiratory infection and antibiotics are not indicated.

Instructions

Step 1: Prepare

Review clinical guidelines for antibiotic stewardship (e.g., CDC or WHO recommendations).

Refresh your knowledge on patient education strategies, including motivational interviewing and shared decision-making.

Step 2: Use AI to simulate the patient conversation

Copy and paste this prompt into your AI tool or chat assistant:

AI Prompt to copy and paste:

"Act as a 42-year-old patient visiting an FNP for nasal congestion, sore throat, and mild cough for four days. You insist on receiving antibiotics because they 'always help' and you don't have time to be sick. You are frustrated and push back against explanations. Respond as this patient, expressing your concerns and resistance."

Engage in a 5–7 minute written or spoken interaction. Focus on:

Validating the patient's concerns.

Explaining why antibiotics are not appropriate.

Offering alternative symptom management strategies.

Maintaining rapport and trust.

Step 3: Write a reflection paper (2–3 pages double spaced)

Your paper should include:

Summary of the interaction: Key points raised by the patient and your main communication approaches.

Self-reflection: How did it feel to decline the patient's request? What was most challenging or surprising?

Application to practice: What strategies worked well to preserve the relationship while providing evidence-based care? What would you do differently in future encounters?

Patient-centered approach: How will you continue to support and educate this patient moving forward?

Tips for use

Encourage use of communication techniques

Students can be prepared to use communication techniques or patient education theories appropriate to the scenario. You can ask the students to prompt the AI to evaluate their communication based on the appropriate technique after completing the role play.

Set expectations for respectful, professional communication

Reinforce that even in simulated scenarios, professional tone and patient-centered communication are essential. Have students share a link to their chat, or copy and paste the chat into a document to submit with their reflection to demonstrate that students are accountable for their role play participation.

Connect to real-world practice

Emphasize how the skills they are practicing in the role play directly transfer to patient care and interprofessional communication. Discuss how the skills support patient safety, trust-building, and effective advocacy.

Normalize iterative improvement

Encourage the students to view the role play as practice, not performance. Emphasize that repetition and iterative feedback are the key to mastering communication skills. Allow students to repeat the role play to test different approaches to the interaction.

Generative AI as a Digital Tutor

A digital tutor can be a powerful solution for educators seeking to extend learning support beyond the classroom, personalize feedback, and engage students in self-directed exploration. By leveraging AI and other digital tools, a digital tutor can offer just-in-time guidance, clarify difficult concepts, and adapt practice activities to individual learning needs. This can help students build confidence and reinforce mastery at their own pace. However, a digital tutor is not a universal fix. It may not be appropriate when learning outcomes rely heavily on human judgment, nuanced communication, emotional support, or other areas where educator expertise and relational teaching are irreplaceable. Additionally, digital tutors are less effective when they are used as a substitute for educator interaction rather than as a supplement, and they do not replace opportunities for students to engage in complex, authentic clinical reasoning or interpersonal skill development. Thoughtful integration, aligned with course objectives and ethical standards, is essential to ensure digital tutors truly enhance rather than dilute the learning experience.

Designing the Digital Tutor

Defining Your Purpose and Scope

Define exactly what you want your digital tutor to accomplish. This will help keep the tutor narrow enough that it can answer student questions based on the materials you select. Consider who your learners are, what support they need most, and how this tool will complement your teaching.

Write a clear instruction

You will write a set of instructions for GPT to define how it can and should behave. Think of it as telling the AI exactly how you want it to "think" and "talk."

Create an example question and response

Write a few example student questions and ideal GPT-style responses. This helps shape how the AI will respond.

Test and Adjust

Paste your system prompt and example questions into a GPT chat (such as a custom GPT builder or a private ChatGPT thread). Try asking different student-style questions to see if it responds as you intend. If it's too detailed, simplify your

instruction. If it misses important steps, add specific directions to the system prompt (e.g., "Always mention safety considerations and common mistakes to avoid").

Share it with students

Once it feels right, share your custom GPT prompt or instructions with students. If available to you, build a custom GPT to share the digital tutor. Tell students the purpose (e.g., practice, review) and how to use it responsibly (not as a substitute for clinical judgment or educator feedback).

Example Prompts

Medication Calculation Tutor

You are a friendly and supportive nursing tutor. You help students understand medication dosage calculations step by step. Always ask the student questions to check their understanding before giving the final answer. Use simple language and encourage the student to explain their thinking.

Example:

Student: "How do I calculate a drip rate for an IV infusion?"

GPT Tutor Response: "Great question! First, can you tell me what total volume you need to infuse and over how many hours? Let's work it out together step by step."

Diabetes Pathophysiology Tutor

You are a friendly and knowledgeable nursing pathophysiology tutor. You help nursing students understand the differences between Type I and Type II diabetes at a cellular and systemic level.

Your goal is to guide students to explain the underlying mechanisms in their own words. You do this by:

- Asking them to describe what they know and correcting misunderstandings gently and clearly.
- Helping them break down complex processes into simple steps.
- Emphasizing the differences in insulin production, insulin resistance, autoimmune factors, and risk factors.

Ask students one question at a time and ensure they can answer correctly before moving on. If the student gives a superficial or partially correct answer, ask them to go deeper. Always encourage students to summarize what they learned at the end of each exchange to reinforce understanding.

Medication Tutor

You are a medication tutor designed specifically for prelicensure student nurses. Your focus is on helping them understand, remember, and safely apply knowledge about one medication category: beta blockers.

Your teaching style is clear, supportive, and clinically practical — no fluff, no patronizing tones. Use simple language when possible. If students answer incorrectly or superficially, ask the question again a different way before explaining key concepts. You can include mnemonics, patient teaching tips, and clinical pearls. Emphasize safety, monitoring, and patient education.

Start the session by asking the student to explain what beta blockers do. Wait for the student to respond before continuing.

Then, one question at a time, ask the student to describe:

- Their general mechanism of action.
- Common indications for beta blockers.
- Key side effects and adverse effects.
- Major nursing considerations (such as when to hold the medication based on vital signs).

Encourage students to ask follow-up questions to test their understanding or clarify points. At the end of the session, provide three multiple choice questions one at time. If the student answers incorrectly, guide them to the correct answer.

AI-Led Mini Simulation

AI-led mini simulations offer nurse educators a flexible way to engage students in realistic decision-making without the logistical barriers of traditional simulation labs. These interactive scenarios allow students to explore reasoning, practice communication, and apply theoretical knowledge in a safe, low-stakes environment. By interacting with AI-generated simulations, students can experience dynamic feedback and immediate consequences based on their choices, which supports deeper learning and retention. Additionally, mini simulations can be easily tailored to diverse clinical contexts, patient populations, or leadership challenges, providing rich opportunities for personalized and equity-focused learning.

Designing a Mini-Simulation

Define the Educational Purpose and Learning Objectives

Decide what skills or competencies you want the students to practice and the level of complexity that is appropriate for the learner. You will need to provide this information in your prompt to ensure the simulation will fit your course goals. The more specific you can be, the better the fit will be.

Select the Scenario Framework

Identify the setting, the main characters, and the central problem or conflict that drives the scenario. Consider the decision points that will be included.

Determine the level of interaction

Decide if students will be given pre-defined options, or if they will type open-ended responses.

Determine the level of randomness

Decide how much freedom you will give the AI to select the next step of the scenario. For example, you may want to define pathways for "correct" and "incorrect" responses from the student. Alternately, you may ask the AI to select the next challenge based on the student decisions.

Determine the level of feedback

Consider whether students will receive feedback from the AI after each decision, feedback from the AI at the end of the simulation, or no feedback from the AI.

Example Prompt

Population Health Emergency Management Simulation

You are a simulation exercise for me. I am a student nurse learning about managing population health in an emergency. The purpose of the simulation is to learn how to prioritize in an emergency. For the simulation, you will assign me a nursing leadership role at an agency or health care organization in Nevada and provide an emergency scenario that requires me to make a series of six decisions.

You will play the role of a leader in the community who would manage the emergency at the city, county, or state level as appropriate. You can use information from publicly available emergency response plans in Nevada to build the scenarios decision points.

Introduce the scenario, inform me of the expected costs and lives lost in this emergency, then provide a decision point where I must choose to prioritize one of the actions. Wait for me to respond to your prompt before moving on. If I make a bad decision, escalate the scenario.

After I select, provide immediate feedback about the results of the decision. The consequences of good and bad decisions may focus on budget issues, staffing issues, resource issues, health issues, or lives lost as appropriate for the situation. When I prioritize an action that makes the situation worse, clearly explain what would have been better but continue the scenario with the bad choice.

After six decisions, provide an update of the situation including total expected costs and lives lost. Provide a debrief about how my decisions contributed to the costs of emergency management, staffing and resource allocation, and health of the community.

Tips for use

Give the AI a specific goal for the simulation, such as a course objective, to keep it focused. If the simulation takes a turn you didn't intend, it may be that the goal is not narrow enough.

Test your prompt in multiple AI tools to ensure students have good experiences whichever AI they use. Alternatively, create a GPT with the prompt and share the link so students can experience it with a single click.

Model an AI simulation for the students so they can see the range of scenarios presented by the AI and the responses that are appropriate for the AI to move the simulation forward.

Focus on the process. AI will not create a flawless simulation. But it can get students thinking, analyzing, and reflecting. Use AI mistakes as teachable moments.

Always include a structured debrief, even if it is asynchronous. Otherwise students could leave with misunderstandings.

If students are completing the simulation outside of class, have them submit a copy of their AI interaction after the simulation. This allows you to review their engagement with the AI and provides an opportunity to debrief based on what students experienced.

Concept Mapping with Generative AI

Generative AI can help student nurses transform an overwhelming flood of information into a visually organized structure. Students can work with a generative AI to brainstorm interconnected topics and temporal relationships, such as physiological changes and responses to interventions. This can help students clarify complex clinical processes and encourage students to engage in deeper critical thinking rather than rote memorization. Used wisely, generative AI can help students build deep understanding.

However, overreliance on AI could prevent independent analysis and contribute to a sense of false mastery if students accept the suggestions uncritically. Educators can consider student readiness for self-directed learning and incorporate appropriate support for AI-collaborative concept mapping. For example, faculty can prepare students to question AI outputs and have strategies to invalidate hallucinations or AI misapplications of information.

Designing a Concept Map Assignment

Consider the level of AI collaboration

Students can use a single prompt that asks the AI to create a concept map. Students can also use a series of questions to generate a list of potential relationships and then test each one for inclusion in a concept map that is drawn by the student. Both methods are valid, but ask the students to apply information in a different way (as an editor or as a co-designer).

Consider the level of student collaboration

Students can build concept maps as individuals or as teams. When students are asked to build concept maps as individuals, they can edit the maps alone or in teams. Alternatively, groups of students can compare maps they each individually co-created with AI, selecting key connections to include in the final map.

Consider error-spotting vs annotation

Educators can create concept maps for students to correct. Students can be asked to identify errors on the map and demonstrate understanding by suggesting an edit. Alternatively, students can be asked to annotate each relationship on the map, identifying if they would keep or delete the connection and providing justification for the decision.

Example Assignment

Purpose

This activity is designed to strengthen your ability to evaluate clinical information, identify relationships, and justify your reasoning. Rather than creating a map from scratch, you will focus on assessing and improving an existing structure — a skill essential for safe nursing practice, where you often critique care plans and identify gaps.

Instructions

Review the educator-provided concept map carefully. Take note of each connection and the flow of ideas.

Annotate directly on the map or in a separate document. For each major connection or branch:

- Explain why the connection is clinically relevant or incorrect.
- Provide supporting evidence (from your textbook, lecture notes, or credible references).

Suggest at least one alternative connection or an additional link you believe should be included.

Identify any areas where you believe the map might be misleading, incomplete, or overly simplified, and explain your reasoning.

Conclude with a brief reflection (approximately 150–200 words) on how this process helped deepen your understanding of the topic and how it might influence your clinical decision-making.

Evaluation Criteria

- Depth and accuracy of annotations.
- Ability to provide evidence-based reasoning.
- Critical identification of potential gaps or errors.
- Thoughtful and clear reflection.

Tips for Use

Provide students (and the AI) with a narrow, focused objective for the concept map. If the AI is given too wide an objective, the concept map will be a jumbled, unreadable mess.

Guide students to check for accuracy. Model how you want the students to respond to mistakes and misinformation in the AI output. Teach them to annotate the output so they stay engaged in the review process.

Scaffold the process by breaking up an assignment into smaller parts. For example, have the students submit a list of relationships for the concept or clinical diagnosis. Then, have the students use that list of relationships to prompt the AI to build a concept map.

AI as a Study Partner

Generative AI offers personalized, on-demand assistance that adapts to individual learning needs, enabling students to grasp complex concepts, reinforce critical thinking, and efficiently manage their study time. Through features such as interactive quizzes and instant feedback, AI promotes active learning and continuous self-assessment. Additionally, it helps students prepare for licensure exams like the NCLEX. By leveraging AI as a study partner, nursing students can strengthen their understanding, improve performance, and build essential skills for modern clinical practice.

While AI offers significant benefits as a study partner, there are important considerations and potential downsides to keep in mind. One key concern is accuracy. AI-generated content may occasionally include outdated or incorrect information, making it essential for students to cross-reference with trusted sources like textbooks, peer-reviewed journals, or faculty guidance. Additionally, overreliance on AI can hinder critical thinking and independent learning, as students might be tempted to accept answers without fully understanding the underlying concepts or connecting multiple pieced of information together. Ethical considerations are also important; using generative AI to complete graded assignments or exams may be considered academic misconduct unless specifically allowed. Furthermore, AI cannot replace the clinical judgment, hands-on skills, and human interaction that are central to nursing practice. Therefore, while AI is a powerful supplement to traditional learning methods, it should be used thoughtfully and in balance with active engagement, mentorship, and clinical experience.

Practical ways students can use AI as a study partner

Clarify Complex Concepts

Students can use AI to simplify and personalize the understanding of challenging nursing topics. This can be achieved by asking AI to explain diseases, procedures, or systems in everyday language or based on your current knowledge level.

Generate Custom NCLEX Practice Questions

Students can use AI to generate NCLEX practice questions to strengthen knowledge retention and exam readiness through active recall and self-assessment. This can be achieved by promoting AI to create targeted practice

questions, include answer explanations and rationales. Students can combine questions to create mini mock exams. Practice questions generation may include misinformation; students are encouraged to verify the answers and use the rationales as the main focus.

Time Management & Personalized Study Plans

Students can use generative AI to organize study time efficiently and prioritize learning based on deadlines and personal goals. To achieve this, students prompt the AI to design a custom study schedule that fits their clinical hours, assignments, and areas needing improvement.

Summarize Notes and Create Flashcards

Generative AI can assist students as they condense large amounts of information for review and convert notes to flashcards that aid in memorization. To achieve this, students will provide the AI with their reading and lecture notes, then prompt the AI to convert the notes content into summaries or flashcards for efficient review.

Generate Study Guides

Students can use generative AI to organize and synthesize their notes into concise, structured guides for exams or topic reviews. To achieve this, students will provide their notes, any class study guides or content outlines into the AI. Then prompt the AI to summarize the notes based on the study guide or content outline. Students can also provide the course or module learning objectives to prompt the AI to reorganize their notes by learning objectives.

Example Prompts

Concept Clarification

Explain heart failure to me like I'm a first-year nursing student.

What's the difference between Type 1 and Type 2 diabetes in simple terms?

Study Questions

Give me 5 NCLEX-style questions at the application level of understanding on fluid and electrolyte imbalances with rationales.

Test me with a 10-question quiz on pediatric nursing interventions focusing on gastrointestinal disorders. Include the rational for each answer choice.

Personalized Study Plans

Make a 3-week study plan to review pharmacology for my final exam. Block out the following times that I have class XX and clinical XX.

Help me schedule study blocks around my clinical shifts. provide clinical shifts]

Summarize Notes

Summarize this paragraph on renal failure into 5 key points.

Turn this list of nursing interventions into flashcards.

Generate Study Guides

Create a study guide for the cardiovascular system including key terms, pathophysiology, and nursing interventions.

Summarize everything I need to know for my exam on infection control into a study guide format.

AI-Generated Nurse Plans

Generative AI can serve as a supplemental educational tool that helps students explore a variety of evidence-based options when developing individualized care strategies or proposing system-level improvements. By inputting realistic clinical or organizational scenarios into AI, students can receive structured guidance that reflects professional standards. Using AI in this way prepares nursing students to be more competent, adaptable, and innovative in both bedside and leadership roles.

To ensure students are practicing the responsibility of providing safe, individualized, and effective patient care, assignments that include generating nursing plans should include assessment and evaluation of the output. AI tools may produce content that is outdated, overly general, or lacking context-specific nuances. By evaluating the AI output, students practice critical thinking, learn to identify and correct errors or gaps, and apply theoretical knowledge to practical situations. This process reinforces learning, prevents the misuse of technology, and helps students develop confidence in their ability to assess and tailor nursing care plans appropriately.

Designing a Nurse Planning Assignment

Break the process into steps

Generative AI is used to support student development of a nursing plan, it does not replace student learning. Single prompts to "create a care plan" will be less effective for learning than multi-step assignments that encourage use of generative AI to suggest possible outcomes, evaluate priorities, and compare intervention options.

Require demonstration of clinical decision making

The AI output should be critically evaluated and students should justify the decisions they make to keep or reject AI suggestions. Design assignments to require the justification for each decision.

Demonstrate adherence to HIPAA standards

Ensure students understand the types of patient data that are considered personal identifiers because inputting protected health information into public generative AI can be a violation of HIPAA. Provide a structured template that helps students prepare prompts that follow HIPAA.

Example Assignment: Nursing Care Plans

Generative AI can assist in care plan development and support learners in exploring nursing diagnoses, identifying appropriate interventions, and structuring care plans in alignment with established frameworks such as North American Nursing Diagnosis Association International (NANDA-I). AI provides instant feedback and suggestions allowing the student to practice self-directed learning and deepening understanding of clinical application. While not a replacement for clinical instruction or evidence-based practice, AI serves as a valuable supplemental tool to support nursing students as they build competence and confidence in care planning.

Scope of Use

AI can be appropriately used to generate sample nursing diagnoses, suggest possible patient outcomes, and recommend interventions based on the information provided about the clinical scenario. AI can also help students understand rationales behind nursing actions and assist in organizing and formatting care plans. Providing a template can help the student view the information in an organized way, if a faculty wishes. This template can be uploaded into AI by the student and prompted to generate output using the template provided.

Instructions for Using AI in Nursing Care Plan Development

1. Clarify Your Learning Objectives
   Begin with clear learning outcomes or care plan requirements. Use your course guidelines and assignment instructions to define what the AI should help you accomplish.

2. Use AI as a Collaborative Tool
   Treat AI as a support resource—like a tutor or brainstorming partner—not a substitute for your clinical judgment or academic work.

3. Input Clear, Specific Prompts
   When interacting with AI, provide complete clinical scenarios, patient data, and your care goals to generate useful and accurate care plan suggestions.

4. Critically Evaluate AI Responses
   Always verify the accuracy and appropriateness of AI-generated content using your clinical knowledge, textbooks, or evidence-based resources.

5. Adapt AI Output to Clinical Standards
   Customize the AI-suggested care plans to align with nursing frameworks.
6. Document Authorship and AI Use Transparently
   Acknowledge use of AI in assignment submissions if required by your instructor or academic policy.
7. Maintain Patient Confidentiality
   Never input real patient names or identifying information into the AI.

Sample Prompts to provide students

Generalized prompt:

Using the following patient scenario (provide very detailed scenario-see below), create a complete nursing care plan. Include: one appropriate NANDA-I nursing diagnosis, 2–3 expected outcomes using NANDA 3–4 nursing interventions from the NANDA, and brief rationales for each intervention. The more detailed the scenario the more detailed and aligned the care plan will be.

Detailed Patient Scenario Examples

Example 1:

"A 72-year-old male patient is admitted to the medical-surgical unit with pneumonia. He has a productive cough, shortness of breath, fever of 101.8°F, and crackles in both lower lung fields. He reports fatigue and difficulty sleeping. His oxygen saturation is 89% on room air, and he is receiving 2L/min of oxygen via nasal cannula."

Example 2:

"A 68-year-old Hispanic female was admitted for heart failure exacerbation. She reports shortness of breath, fatigue, and swelling in her lower legs. Vital signs: BP 150/90, HR 98, RR 24, O2 sat 90% on room air. She has a history of hypertension and diabetes, lives alone, smokes daily, and reports anxiety about her health. Labs show elevated BNP and mild hyponatremia. She is at high fall risk and needs Spanish-language educational materials."

Specifically focused prompt

"Create a complete nursing care plan for a 45-year-old male patient admitted with acute appendicitis post-appendectomy. He is experiencing moderate abdominal pain, has a surgical incision with slight redness, and reports anxiety about recovery. Include one appropriate NANDA-I nursing diagnosis, 2–3

expected outcomes (NOC), 3–4 nursing interventions (NIC), and rationales for each intervention. Focus on pain management and prevention of post-operative complications."

Analysis and editing of AI Output

After AI has generated a nursing care plan using an effective AI prompt and detailed case scenario information, verify and critically analyze the output. Analysis should include verification for accuracy, relevance to the patient scenario, alignment with current evidence-based practices, and consistency with standardized nursing frameworks. Edit and make changes directly on the document.

Example Assignment: Quality Improvement Plans

AI can assist by generating ideas based on best practices, helping students structure QI projects using established frameworks such as PDSA (Plan-Do-Study-Act), and offering guidance on measurable outcomes, root cause analysis, and intervention strategies. By simulating real-world clinical challenges and proposing data-informed solutions, AI encourages critical thinking and systems-based reasoning. It also allows students to explore innovative approaches to common problems such as medication errors, fall prevention, or communication breakdowns. Ultimately, integrating AI into the QI planning process equips future nurses with the skills to lead change, promote evidence-based practice, and actively contributes to a culture of continuous improvement in healthcare settings.

Scope of use

AI can be used to assist students in idea generation, identifying clinical problems, outlining goals, proposing interventions, project structuring, strategy refinement and suggesting metrics for evaluating success. It can also help frame QI projects using established models such as PDSA (Plan-Do-Study-Act), SBAR (Situation-Background-Assessment-Recommendation), or Root Cause Analysis. Students are expected to practice ethical use, avoiding the inclusion of any protected health information (PHI) and clearly documenting any AI assistance. Within these boundaries, AI offers a valuable resource to help students learn how to design, implement, and evaluate effective QI initiatives in nursing practice.

Instructions for Using AI in Quality Improvement Plan Development

1. Define the Purpose of Your QI Project
   Clearly identify the clinical issue or system-based problem you want to improve (e.g., medication errors, hand hygiene compliance, communication breakdowns). This will guide the AI in generating relevant and focused responses.
2. Use Specific, Structured Prompts
   When interacting with AI, provide detailed context. Include the clinical setting (e.g., inpatient unit, outpatient clinic), the issue at hand, and your goals. For example:
   "Help me develop a QI plan using the PDSA model to reduce patient falls on a medical-surgical unit."
3. Request Support for QI Components
   Ask the AI for assistance with key elements such as:
   a. Problem statements and background
   b. SMART goals (Specific, Measurable, Achievable, Relevant, Time-bound)
   c. Root cause analysis ideas
   d. Intervention strategies
   e. Outcome measures and evaluation methods
   f. Implementation steps and timelines
4. Critically Evaluate AI Suggestions
   Review AI-generated content for clinical accuracy, feasibility, and alignment with evidence-based practice. Cross-check against nursing literature, guidelines, and organizational policies.
5. Incorporate Evidence and Scholarly Sources
   Use AI to brainstorm ideas but always support your QI plan with peer-reviewed research, national standards (e.g., QSEN, IHI), and institutional best practices.
6. Adapt AI Output to Fit Real-World Application
   Customize the generated plan to match the resources, culture, and needs of your chosen healthcare setting. AI should only assist with structure and ideas.

7. Maintain Academic Integrity and Confidentiality
   Do not include real patient identifiers, protected health information (PHI), or private organizational data. Cite your use of AI tools in your assignment or project documentation.
8. Use AI as a Learning Partner—Not a Final Authority
   Let AI serve as a springboard for deeper thinking, creativity, and learning. Always rely on your nursing judgment, faculty input, and evidence-based guidelines when finalizing your QI plan.

Sample Prompts to provide students

Example 1:

"Develop a QI plan focused on improving medication administration accuracy in a long-term care facility. Use evidence-based strategies and include a problem statement, one or two SMART objectives, contributing factors (such as distractions, incomplete documentation, or staffing shortages), and at least three nursing-led interventions. Include measurable indicators for tracking progress (e.g., error rates, incident reports) and explain how nurse education and workflow changes could help reduce errors. Structure the plan using the PDSA model."

Example 2:

"Create a nursing quality improvement (QI) plan using the PDSA (Plan-Do-Study-Act) model to reduce inpatient falls on a medical-surgical unit. Include a clear problem statement, SMART goals, possible root causes, proposed nursing interventions, and measurable outcomes. The plan should focus on improving staff awareness, patient safety protocols, and hourly rounding practices. Also, suggest methods to evaluate the effectiveness of the intervention over a 3-month period."

Analysis and editing of AI Output of a QI Plan

While AI can generate helpful frameworks and ideas, it may also produce vague, overly broad, outdated, or incorrect content. Students use clinical reasoning to evaluate whether the AI output meets the requirements of the assignment and reflects current best practices. By reviewing and editing the AI-generated QI plan, students reinforce their understanding of QI methodologies.

Student Instructions for Editing AI Output for a QI Plan – Step-by-Step

1. Check for Accuracy and Relevance

- Confirm that the clinical issue is clearly defined and based on actual nursing concerns.
- Ensure that all suggested interventions are appropriate for the setting described (e.g., hospital vs. long-term care).

2. Refine the Problem Statement
    - Make it specific and measurable.
    - Example: Change *"There are too many falls"* to *"The medical-surgical unit experienced a 20% increase in patient falls over the past quarter, exceeding the hospital's benchmark by 8%."*
3. Evaluate and Rewrite SMART Goals
    - Ensure goals are Specific, Measurable, Achievable, Relevant, and Time-bound.
    - Example: Instead of *"Reduce falls,"* revise to *"Reduce patient falls on the med-surg unit by 15% within three months through the implementation of hourly rounding and staff education."*
4. Customize Interventions
    - Adapt suggestions to the facility's resources and staffing.
    - If AI suggests "increase nurse staffing," consider modifying it to "restructure staffing assignments during peak hours to reduce patient wait times for assistance."
5. Ensure Evidence-Based Support
    - Verify the interventions align with guidelines from sources like the Institute for Healthcare Improvement (IHI), QSEN, or The Joint Commission.
    - Look up current research to justify your strategy. Verify that any evidence cited by AI actually exists and supports your strategy.
6. Strengthening the Evaluation Plan
    - Make sure the outcome measures are clear and quantifiable.
    - Example: Instead of *"Evaluate fall reduction success,"* use *"Track and compare the number of fall-related incident reports monthly for three months."*